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A Numerical Investigation of Scalar Inflation Fields in the Standard Model of Cosmology

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Abstract

In this report we review how the Friedmann equation can be derived from the Friedmann-Lemaître-Robertson-Walker (FLRW), which then relates the Hubble parameter to energy density, spacetime curvature, and the cosmological constant. We solve the Friedmann equation analytically for several single fluid energy density approximations and numerically using Python for a mixed energy density model. This model can be shown to be consistent with observation. We discuss how observations constrain this model to require fine tuning in initial conditions, and show how inflationary cosmological models provide an explanation for this fine tuning. We then explore the scalar field equations that can produce the required inflation, and we numerically solve the equations for various scalar field potentials using Python, and demonstrate their attractor properties. We focus on six potentials and discussed how Planck 2018 Cosmic Microwave Background (CMB) anisotropies impose constraints on inflationary models, with the Starobinsky model being the most applicable in light of these observations. We relate this discussion to the search for a theory of Quantum Gravity and the need for an inflationary model that aligns with certain properties.

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1 Introduction

The development of science is characterised by a perpetual battle of attrition between theoretical development and, more often than one would like, conflicting observational evidence. Frequently, the seemingly intractable gulf between what is understood and what is observed, is bridged by a flash of insight by those individuals whose perspective illuminates the baseless, yet seemingly obvious, assumptions previously made about the nature of the universe whether it be that the earth is the centre of the universe (Debus [2004]), that space and time itself is absolute and unchanging (DiSalle [2008]), or that objects even exist in some definite state when they aren't measured (Mermin [1985]). These novel perspectives propel the field into unmapped landscapes, and provides science with a firmer foundation for future theories. Nowhere has this been more apparent than in the transformation of physics in the turn of the 20th century and it's development in the following decades.

The publication of Einstein's original and elegant field equations (1.1) in 1915 replaced Newton's theory as the most precise description of gravity. By employing the newly developed framework of Riemannian geometry, it was able to achieve a greater level of precision in explaining the effects of gravity in extreme situations involving high speeds or massive objects, where Newtonian theory would later prove to be inadequate. It could explain the flower-like shape Mercury traced out in its orbit around the Sun, where Newton's theory could only see an ellipse (Einstein [1920]), but it also had deep implications for the nature of space and time. It replaced the flat, absolute stage of Newton with a 4-dimensional manifold, one free to curve and warp in the presence of matter.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1.1}$$

In Eq. 1.1, as summarised in Misner et al. [1995] : "Space [$G_{\mu\nu}$] tells matter how to move, Matter [$T_{\mu\nu}$] tells space how to curve". More precisely, the Einstein tensor $G_{\mu\nu}$ describes the geometric curvature of the 4-dimensional "stage" of space-time due to presence of objects within it. This curvature of space-time

in turn alters the trajectory of these objects (described by the energy-momentum tensor $T_{\mu\nu}$), just as a valley or hill alters the path of a marble rolling through it. The energy-momentum tensor describes the distribution of energy and momentum in spacetime. It contains information about the energy density, momentum density, and stress of a physical system. These two terms are related to each other in a proportion given by the constant $\frac{8\pi G}{c^4}$. The Einstein Tensor is made up from the Ricci tensor and Ricci Scalar:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \tag{1.2}$$

which are both in turn made up from the spacetime metric and its derivates. In fact, the Ricci scalar simply is the trace - or sum of the diagonal components - of the Ricci Tensor:

$$R \equiv R^{\mu}_{\mu} \tag{1.3}$$

This spacetime metric is a tensor that describes how distances and angles are measured in a curved space, and both the Ricci tensor and scalar use this to measure how much spacetime is distorted in a given region. In General Relativity, free-falling objects travel along geodesics, which are the shortest paths between two points through curved spacetime. Put simply, as spacetime curves around massive objects, geodesics, despite being the shortest possible distance through space time, seem to mysteriously curve towards massive objects. In other words as we move forward through time, things fall towards heavier objects. This is gravity.

The key property Einstein used to guide his search for the field equation, was the fact that the covariant derivative for both sides of the equation should be zero - a fact which stemmed from the Bianchi Identy (Misner et al. [1995]). Indeed, this line of reasoning led him away from his initial incorrect assumption that the left side of Eq. 1.1 should just be the Ricci Tensor, which did not have a covariant derivative of zero, before he correctly used $G_{\mu\nu}$ (which has the additional $Rg_{\mu\nu}$ term), which did. However, further adding an additional $\Lambda g_{\mu\nu}$, also leaves the covariant derivative of the left side of Eq. 1.1 equal to zero, which implies that the most general form of the Einstein Field equations should contain this additional term. The physical interpretation of such a term, is that it describes the effective energy density of empty space, it manifests as a background force intrinsic to space itself, that produces an accelerated expansion or contraction of empty space. It has no effect on local dynamics, but has the power to determine

that large scale evolution of any universe's cosmology. Einstein initially, and mistakenly, used the comsological constant to ensure his equations would describe a stationary universe, but when Hubble discovered that space was indeed expanding, Einstein abandoned it (O'Raifeartaigh et al. [2017]). Many decades after his death in 1998, forseen by practically no one, two seperate projects independently made the discovery that the expansion of the Universe was actually accelerating (Peebles and Ratra [2003], Goldhaber and Cline [2009], nob [2011]). Λ was raised from the dead, now with the goal of doing the opposite of what it initially was made for. In the current understanding of cosmology, it is the manifestation of "Dark Energy", a background repulsive force intrinsic to space itself, that counteracts the attractive force of gravity to produce an accelerated expansion of empty space (Liddle [2015]). As of yet, there is currently no accepted explanation of its cause (Liddle [2015]).

The deceptively concise tensoral form of the field equations conceals the fact that it is in fact a system of 16 coupled non-linear partial differential equations. The Einstein tensor itself, $G_{\mu\nu}$, has 10 independent components in 4-dimensional space. The deep beauty of the Field equations is inseperable from the difficulty of their solutions.

1.0.1 Derivation of Friedman Equations

Three years before his early death, Alexander Friedmann found an exact solution to the Field equations (Tropp et al. [2006]). He was not the first, six years earlier during World War 1, on the opposite side of the Russian front to Friedmann, Karl Schwarzschild had miraculously discovered his eponymous metric only a few months after Einstein's publication, and with it he become the first to glimpse the existence of black holes (Schwarzschild [1992]). But while Schwarzschild wouldn't make it out the war, Friedmann survived to develop his cosmological solution to the Field equations, before he too succumbed to mortal illness. On the largest scale, individual galaxies become as insignificant as atoms in a cup of water, and the universe appears flat and uniform in all directions and locations (Liddle [2015]. At this scale, all matter and radiation in the universe can be approximated as a homogeonous isotropic fluid, described by the diagonal energy-momentum tensor:

$$T^{\mu\nu} = \begin{pmatrix} \rho(t) & 0 & 0 & 0 \\ 0 & -p(t) & 0 & 0 \\ 0 & 0 & -p(t) & 0 \\ 0 & 0 & 0 & -p(t) \end{pmatrix} \tag{1.4}$$

Note the identical spatial components. The corresponding metric that describes the space-time in which this uniform fluid exists is given by multiplying a static spatial geometry with a time-dependent scale factor a(t)

$$ds^2 = dt^2 - a^2(t)d\vec{x}^2 \tag{1.5}$$

which in turn corresponds to the following metric:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -a(t)^2/(1-kr^2) & 0 & 0 \\ 0 & 0 & -a(t)^2 r^2 & 0 \\ 0 & 0 & 0 & -a(t)^2 r^2 \sin^2\theta \end{pmatrix} \tag{1.6}$$

Now it is possible to compute every component of the Ricci tensor (and thus according to Eq. 1.3 the Ricci Scalar also) though we will not explicitly do so here.

From Eq. 1.6 one will find the Ricci scalar to equal

$$R = -6\frac{1}{c^2 a^2}(a\ddot{a} + \dot{a}^2 + kc^2) \tag{1.7}$$

We substitute the R_{00} component of the Ricci Scalar back into the Einstein field equations along with all the other 00-components of the rest of the tensors. Here we use Eq. 1.1 with the explicit form of $G_{\mu\nu}$ shown in Eq. 1.2:

$$R_{00} - \frac{1}{2}Rg_{00} - \Lambda g_{00} = \frac{8\pi G}{c^4}T_{00} \tag{1.8}$$

Subsituting in the components from Eq. 1.6, the subsequent calculation of R_{00} from $g_{\mu\nu}$, Eq. 1.7, and Eq. 1.4, we find

$$-\frac{3}{c^2}\frac{\ddot{a}}{a}\frac{a}{a} + \frac{3}{c^2}\frac{1}{a^2}\left(a\ddot{a} + \dot{a}^2 + kc^2\right) - \Lambda = \frac{8\pi G\rho}{c^2} \tag{1.9}$$

With some trivial algebraic manipulation and healthy motivation, one can derive the first Friedmann equation (Misner et al. [1995]):

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3} \tag{1.10}$$

The scale factor a represents the relative size of the universe compared to its size at some initial time, and it is closely related to the cosmological redshift, that is, how much the wavelength of light from a distant source has been stretched due to the expansion of the universe. As the universe expands, the scale factor increases, and the distances between galaxies also increase. $H \equiv \frac{\dot{a}}{a}$ is the Hubble parameter and it measures the rate of expansion. A positive value of the Hubble parameter, $H > 0$, describes an expanding universe while a negative Hubble parameter, $H < 0$, describes a collapsing one (Kinney [2009]). Eq. 1.10 reveals that rate at which the scale factor changes is determined by the energy content of the universe, which in turn depends on the content of matter and radiation contained by it, as well as the value of the cosmological constant.

Now, using the spatial components of the tensors in Eq. 1.1, one can calculate the trace of each tensor in the Field Equations. We do so by applying the contravariant form of the metric to each tensor (as is done with Eq. 1.3):

$$R_{\mu\nu}g^{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}g^{\mu\nu} - \Lambda g_{\mu\nu}g^{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}g^{\mu\nu} \tag{1.11}$$

We use Eq. 1.3 to simplify the first term, and since $g_{\mu\nu}g^{\mu\nu} = \delta^{\mu}_{\mu}$, which in turn is equivalent to taking the trace of the 4x4 identity matrix, we find

$$-R - 4\Lambda = \frac{8\pi G}{c^4}T^{\mu}_{\mu} \tag{1.12}$$

Once again, into Eq. 1.12 we substitute in Eq. 1.7 and the the trace of Eq. 1.4 to find

$$-\left(\frac{-6}{c^2a^2}\left(a\ddot{a} + \dot{a}^2 + kc^2\right)\right) - 4\Lambda = \frac{8\pi G}{c^4}(\rho c^2 - 3\rho) \tag{1.13}$$

Intimidating, though thankfully this simplifies to the much more approachable second Friedmann equation, sometimes referred to as the acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) \tag{1.14}$$

Many may be familiar with the popular statement “Energy cannot be created or destroyed". In a way, this can be seen as another consequence of the Bianchi Identity alluded to earlier. Mathematically this manifests as the fact that the covariant derivative of Eq. 1.4 is equal to zero. This is known as the conservation of the Energy-Momentum tensor $T_{\mu\nu}$, which yields the relation:

$$\dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) = 0 \tag{1.15}$$

which describes the evolution of the energy density over time. With these equations, Friedmann described the kinematics of such a universe. It revealed a universe that was neither static nor eternal, but one that could expand and evolve over time, challenging assumptions held by astronomers since the 16th century, and Einstein himself.

Now that the paralysing complexity of the Field Equations has been tamed to a far more manegeable pair of coupled odinary differential equations (ODEs) in this specific case of a homogeonous and isotropic universe (a condition which ours thankfully obeys), the equations are now susceptible to being solved analytically, as well as through numerical means.

2 Friedmann Equations

2.0.1 Single fluid solutions to the Friedmann Equations

We will now explicitly solve the equations for several single fluid approximations: in the case of a universe dominated by matter (2.0.1), in the case of a universe dominated by radiation (2.0.1), and in the case of a universe dominated by the cosmological constant (2.0.1) - first through analytic means, and then via a numerical method. In all of these cases, we begin by assuming the expansion of the universe can be wholly attributed to the energy density term, discarding all other terms which we assume to be negligible in their contribution to the scale factor (in fact, observations have constrained k - the curvature of the observable universe - to be extremely and suspiciously close, if not equal, to zero (and N. Aghanim et al. [2020]). We discuss the implications of this fact in more detail in Section 3). We only concern ourselves with the following relation:

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho \tag{2.1}$$

which is just Eq. 1.10 without the curvature and the cosmological constant term.

In order to solve Eq. 1.14 or Eq. 1.15, we require an equation of state to relate the pressure in terms of energy density. The general form of such an equation for a general homogeneous, isotropic mass-energy is given by

$$p = \omega\rho \tag{2.2}$$

Matter

Matter, refers to non-relativistic material (such as planets, nebula, stars $etcetera$), which exerts negligible pressure. In Eq. 2.2 we can then simply set the equation of state constant $\omega = 0$ to yield

$$p = 0 \tag{2.3}$$

This is a good model of a cold universe, consisting only of atoms which rarely collide or interact. When we substitute Eq. 2.3 into Eq. 1.15, we obtain the following:

$$\dot{\rho} = -3\left(\frac{\dot{a}}{a}\right)\rho \tag{2.4}$$

This is a separable ordinary differential equation of the first order, and when rearranged and integrated yields the integral equation:

$$\int \frac{d\rho}{\rho} = -3\int \frac{da}{a} \tag{2.5}$$

Thus, it is found that

$$\rho \propto \frac{1}{a^3} \tag{2.6}$$

In words, density decreases as the volume of the universe increases. This result is expected and agrees with intuition. If a box enclosing a fixed number of particles, or a constant mass, were to increase in size, the ratio of the matter with the size of the box decreases. When Eq. 2.6 is substituted Eq. 2.1, we acquire another first-order ODE which we solve either by separation of variables and integration or instead by use of an *ansatz*. Thus, with appropriate integration conditions, the time evolution of the energy density; of the scale factor; and of the Hubble parameter H, is respectively

$$a(t) = \left(\frac{t}{t_0}\right)^{2/3} \tag{2.7}$$

$$\rho(t) = \frac{\rho_0}{a^3} = \frac{\rho_0 {t_0}^2}{t^2} \tag{2.8}$$

$$H \equiv \frac{\dot{a}}{a} = \frac{2}{3t} \tag{2.9}$$

Here, t_0 refers to the current age of the universe (specifically in the rest frame of the Cosmic Microwave Background, discussed in more detail in Section. 2.0.3), ρ_0 is the currently observed value of the energy density, and we use the popular convention of setting the current value of the scale factor $a_0 \equiv 1$.

We plot the analytic solutions below:

Figure 2.1

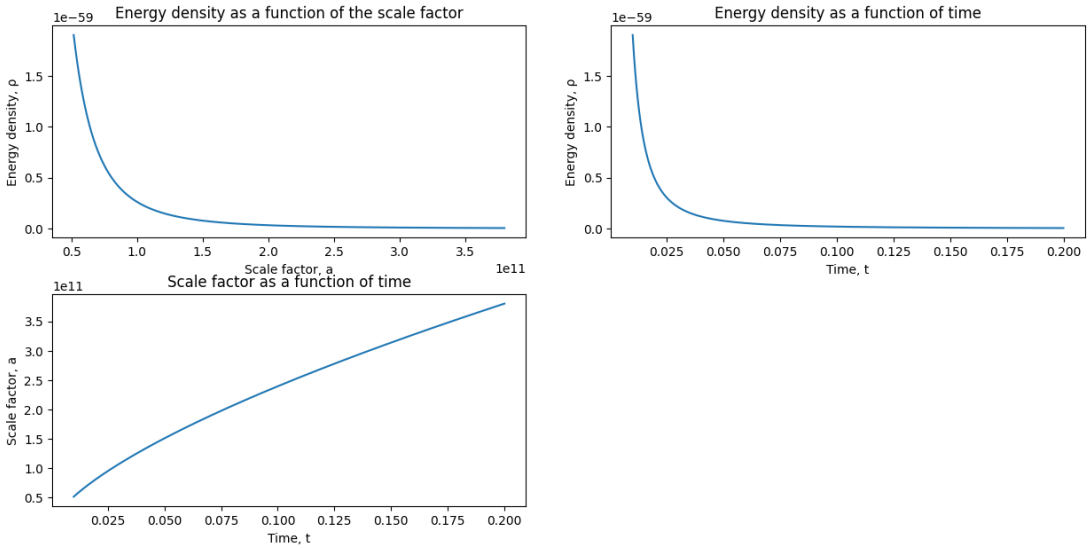


Figure 2.1: Plot of analytic solutions for a universe composed of dust.

Just from inspection of Eq. 2.6, one can see that in the top left plot of Figure 2.1, depicting $\rho(a)$ or the energy density as a function of the scale factor is what we expect. The same is also true of the bottom left graph, and top left graph, from inspection of Eq. 2.8 - or more simply the general relations $\rho(t) \propto \frac{1}{a^3}$ and $\rho \propto \frac{1}{t^2}$ respectively. Note that for these analytic plots, we have truncated the time displayed, in order to emphasise the general shape of the curve.

If we use the expression for $H(t)$ found in Eq. 2.9 to estimate the age of the universe, however, even using the conservatively large value of Hubble's constant found in Riess et al. [2019] - 74.03 +/- 1.42 km/s/Mpc - we find that apparently the current age of the universe is given by $t_0 \approx 9.6$ Gyrs. This figure blatantly disagrees with measurements of the age of globular cluster NGC 6397, 13.4 ± 0.8 Gyr, found by Pasquini et al. [2004]. Which implies, therefore, there must have been another era - or several - within which expansion was dominated by some other fluid.

Numerical solutions for Matter

We also used a numerical method for solving the equations, described in detail in Section 2.0.1.

We first solve backwards in time from t_0 to yield the following plots:

Figure 2.2

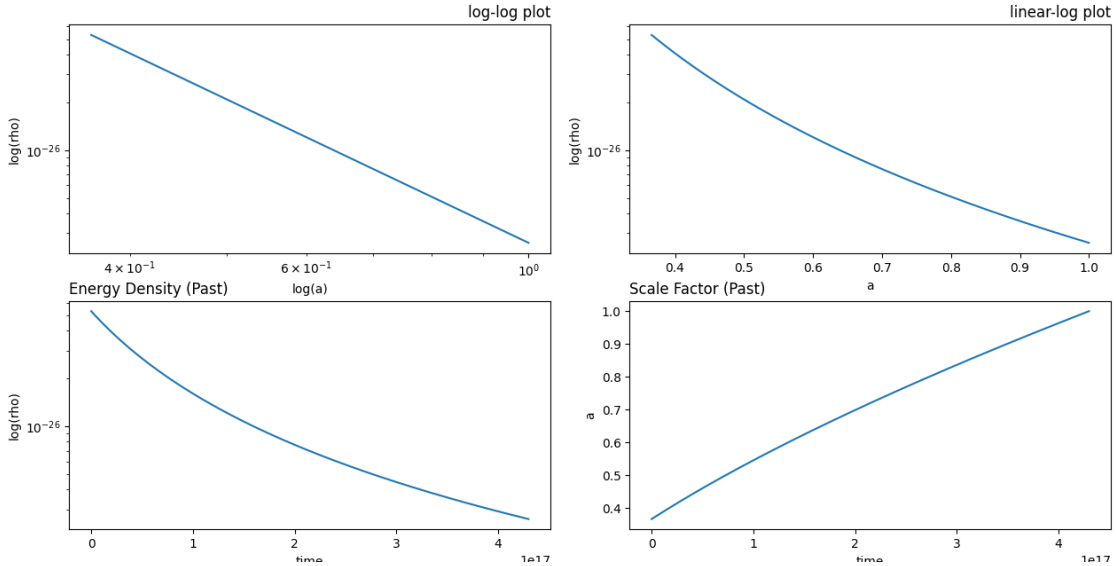


Figure 2.2: Negative time evolution of the universe, from the present at t_0 into the past, for universe composed of matter. To produce these plots, we solve the single fluid equations backwards in time into the past, from the present day. Note that these solutions are entirely equivelent to those attained from analytic calculation, in Fig. 2.1, aside from scaling. Here, the bottom two plots are depict the past time evolution of the energy density ρ and the scale factor a respectively, whilst the upper two plots depict the energy density as a function of rho - as in Eq. 2.6 - with alternate scalings. The upper right plot depicts the linear relationship of the energy density ρ versus the scale factor a, as expected from inspection of the equation.

We then solve forwards in time, essentially producing revered images of the plots found in Fig. 2.2.

Figure 2.3

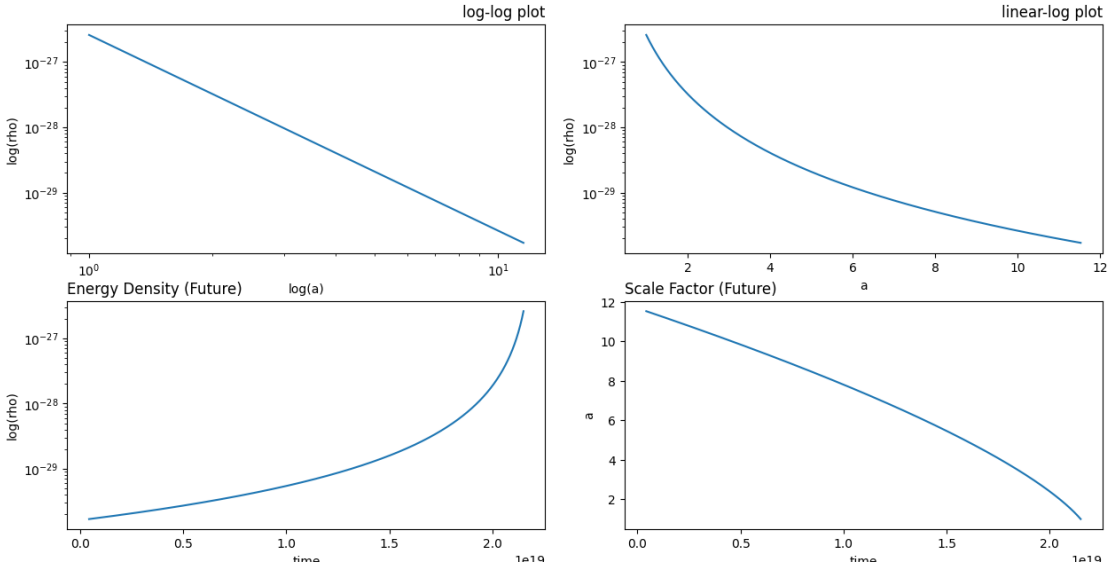


Figure 2.3: Positive time evolution of the universe, from the present at t_0 into the future, for universe composed of matter. To produce these plots, we solve the single fluid equations forwards in time into the future, from the present day. These solutions are also equivelent to those attained from analytic calculation, in Fig. 2.1. Once again the top right plot depicting the future time evolution of the energy density ρ as a function of the scale factor a, with a logarithmic scaling, exhibits an obvious linear dependency as expected. Also note the decay of the rate of expansion for a universe dominated by matter in the bottom right plot, depicting a as a function of time.

Radiation

Radiation deals with particles travelling at relativistic speeds such as light, and neutrinos. It offers a good approximation of the early universe where radiation played a much more dominant role in the expansion relative to today. Unlike matter its pressure is not negligible, instead $\omega = \frac{1}{3}$ and so Eq. 2.2 yields $p = \frac{\rho c^2}{3}$. Substituting this into Eq. 1.15, we see that

$$\dot{\rho} = -4\left(\frac{\dot{a}}{a}\right)\rho \tag{2.10}$$

Once again we rearrange and integrate to obtain the integral equation:

$$\int \frac{d\rho}{\rho} = -4\int \frac{da}{a} \tag{2.11}$$

Thus, in the case of a radiation dominated universe, the time evolution is given by the following relations:

$$a(t) = \left(\frac{t}{t_0}\right)^{1/2} \tag{2.12}$$

$$\rho(t) = \frac{\rho_0}{a^4} = \left(\frac{t}{t_0}\right)^{2/3} \tag{2.13}$$

$$H \equiv \frac{\dot{a}}{a} = \frac{1}{2t} \tag{2.14}$$

Plotted analytically, the first two relations yield:

Figure 2.4

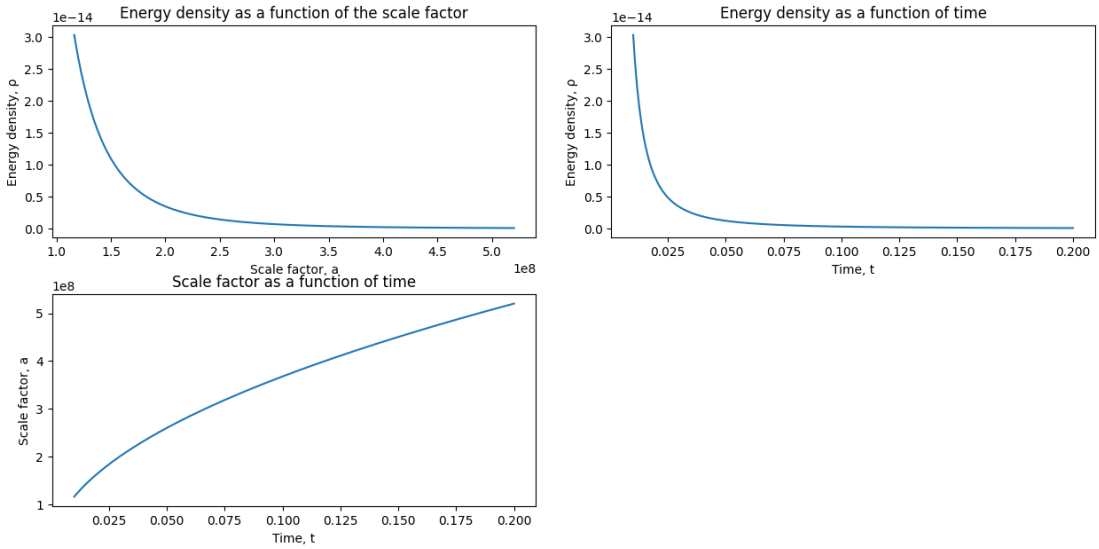


Figure 2.4: Once again, we have truncated the time displayed to emphasise the exponential structure of the curve. As expected from the qualitative consideration of the additional factor of cosmological scaling on the energy density of radiation discussed below, we see in the first to curves a far more dramatic increase in ρ for smaller values of a, compared with Section 2.0.1.

We see that the energy density of radiation drops off at a faster rate than of matter, as it decreases in energy density due to two factors. Firstly, as a result of its dilution as total volume of the universe grows, and secondly as a result of the stretching of its wavelength due to the expansion of space - as a massless particles energy is proportional to its wavelength, as given by Planck's famous relation:

$$E = h\nu = h\frac{c}{\lambda} \tag{2.15}$$

Where E is the energy of a photon, h is Planck's constant, ν is the frequency of a photon, c is the speed of light, and λ is the wavelength of a photon. As

space expands, so does λ, conversely ν. In a sense, the expansion of space saps the energy of any photon travelling through it. The implications of this is that radiation had a large influence at early stages of the Universe's evolution, where space was smaller and hotter, though its rapid decay soon led to an era of matter dominated expansion

Numerical solutions for Radiation

Once again we utilise the method in Section 2.0.1 as an alternate method for solving and plotting these relationships between variables. Firslty, we solve the equations backwards in time into the past:

Figure 2.5

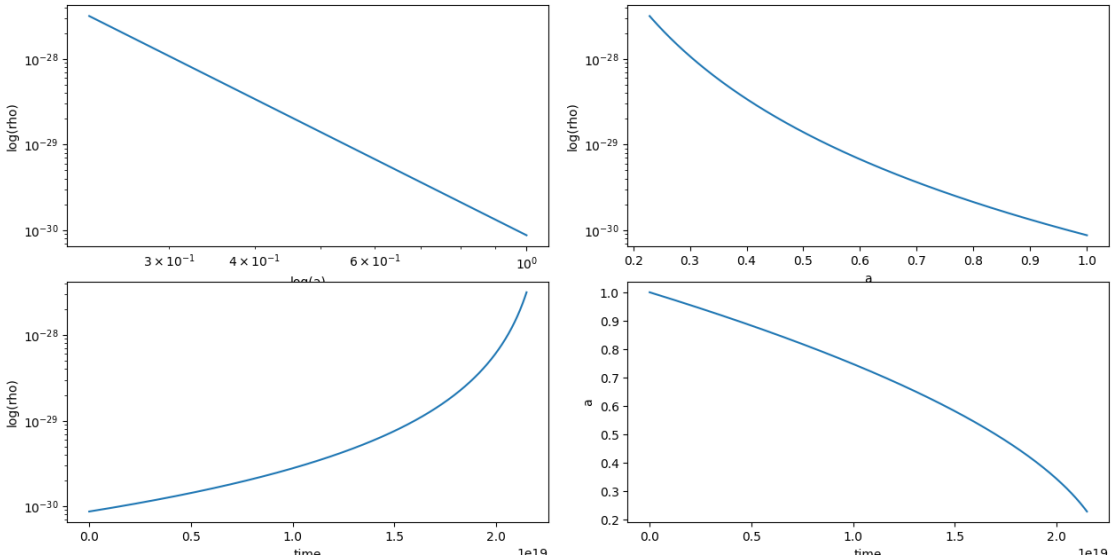


Figure 2.5: Negative time evolution of the universe, from the present at t_0 into the past. To produce these plots, we solve the single fluid equations backwards in time into the past, from the present day. We contain an additional sub-plot relative to Fig. 2.4, the top left sub-plot - that logarithmically scales both axis in the graph of ρ as a function of a. As in the case of Section 2.0.1, these solutions are entirely equivalent, regardless of the method to attain them. The superposition of their plots appear mutually indistinguishable from the original plots.

Followed by a forward time evolution into the future:

Figure 2.6

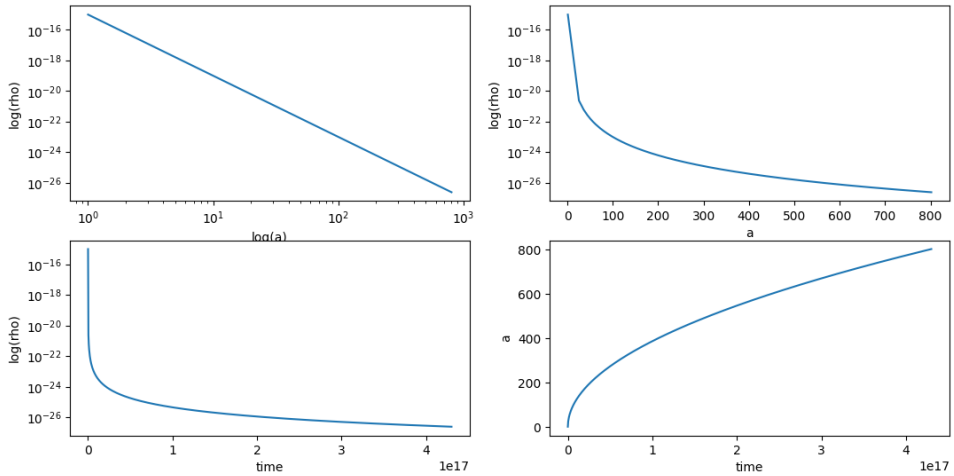


Figure 2.6: Positive time evolution of the universe, from the present at t_0 into the future. The plots on the left side of the figure depict the energy density ρ's rapid decay with respect to both time and scale factor, whilst the bottom right plot shows the consequence on the evolution of the scale factor: the rate of increase a's decreases to approaches a constant value, which manifests as a near linear relation on the rightmost side of the plot.

Cosmological Constant

The method of solving the Friedmann equations for the case of a Cosmological Constant dominated expansion of the Universe diverges somewhat from the previous equations. In the case of the cosmological constant, through a mathematical sleight of hand, we can also treat the influence of Λ as an effective energy density in its own right. From Eq. 1.4, we see that if $T^{\mu\nu} = \Lambda g^{\mu\nu}$ then the corresponding form of Eq. 2.2 is simply given by the following constant ω = -1.

$$p_\Lambda = -\rho_\Lambda \tag{2.16}$$

Substituting this into Eq. 1.15 simply gives us the fact that

$$\dot{\rho_\Lambda} = 0 \tag{2.17}$$

or that the energy density is constant. The physical implication of this is that as the universe expands, so does the vacuum energy density, thus further driving expansion, causing accelerated expansion. We solve Eq. 1.10 for a universe in which the cosmological constant dominates, known as a der Sitter universe

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{c^2}{3}\Lambda \tag{2.18}$$

We seperate and integrate once again

$$\int \frac{da}{a} \propto \sqrt{\Lambda}\int dt \Rightarrow \ln a \propto \sqrt{\Lambda}t \tag{2.19}$$

We can finally see that this is equivalent to

$$a \propto e^{\sqrt{\Lambda}t} \tag{2.20}$$

We visually see these relationships with the following plots

Figure 2.7

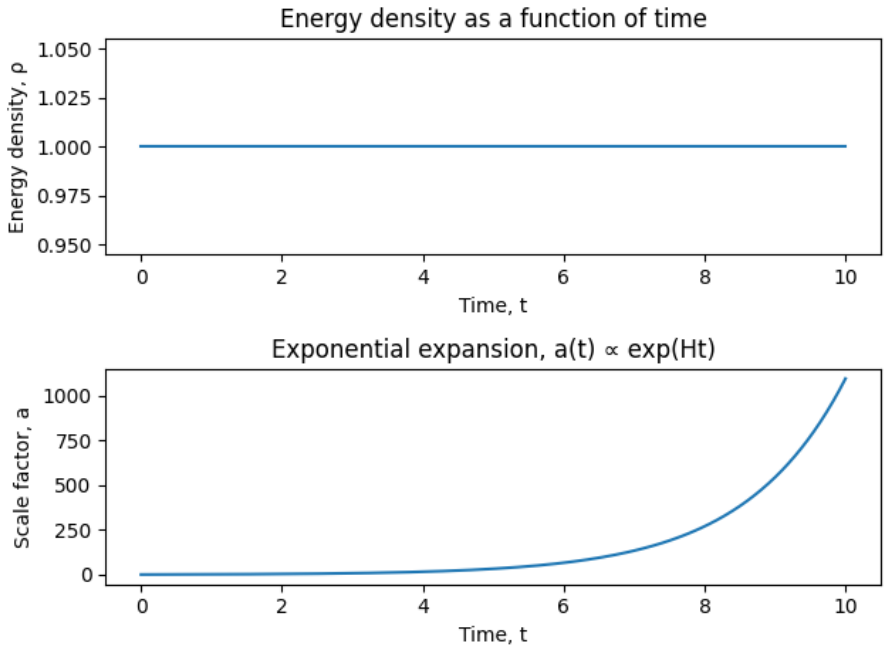


Figure 2.7: As expected for a der Sitter universe, the energy density remains constant throughout its evolution, causing the scale factor to exponentially diverge as described previously. It is worth noting that despite the use of real physical constants in the analytic plots, a der Sitter universe is purely a toy model to demonstrate the contribution of vacuum energy to the behaviour of the universe's expansion, and thus precise numerical values do not convey a significant physcial meaning, especially in the absence of normalising factors as in the proportionality relations utilised.

These solutions are well understood, but they provide a good check for the validity of the numerical method, which will prove useful in the following scenario in which there is no easy way to calculate analytic solutions.

The Numerical Method

We numerically solve the Friedmann equations, first by defining equations in Python, and then by creating a time array to integrate the equations stepwise, either into the past or future.

Specifciallym the general structure of the Python code that would allow a first order ODE to be solved numerically, given some set of initial values, and a time array with which to be integrated with respect to, is given firstly by an initial block of code that defines the functions - that take in as an argument an array containing the initial values of the values of the variables in question - as well as the ODE of interest. Then the arrays are constructed in the following lines, utilizing constants packaged inside some variables, defined earlier on in the program for conciseness. Then an array is set, using the "odeint()" function from the "SciPy.integrate" sub-package. This function will call the functions containing the initial array, time array, and ODE, and iteratively integrate the function for each step in the time array, storing each result in an output array. The columns in the output array correspond to each variable in the ODE that has now been integrated with respect to time. The final block of code merely utilizes the MatPlotLib.PyPlot to produce various plots.

In the specific case for solving the Friedmann equation, the variables processed by the program were the scale factor, and energy density for 3 different perfect fluids: dust/matter, radiation, and the cosmological constant. The Friedmann equation was solved and plotted both forwards and backwards in time.

2.0.2 Mixed Fluid Solutions

Whilst these solutions give suitable approximations of our universe for a given period of its history, a more accurate model describing the evolution of our universe would contain a mixture of fluids, specifically a superposition of all 3 previous fluids investigated. Yet an analytic approach to solving such an equation becomes extremely difficult, if not impossible, for 3 different fluids. We must rely on the numerical methods.

The plots obtained from the numerical method displayed below:

Figure 2.8

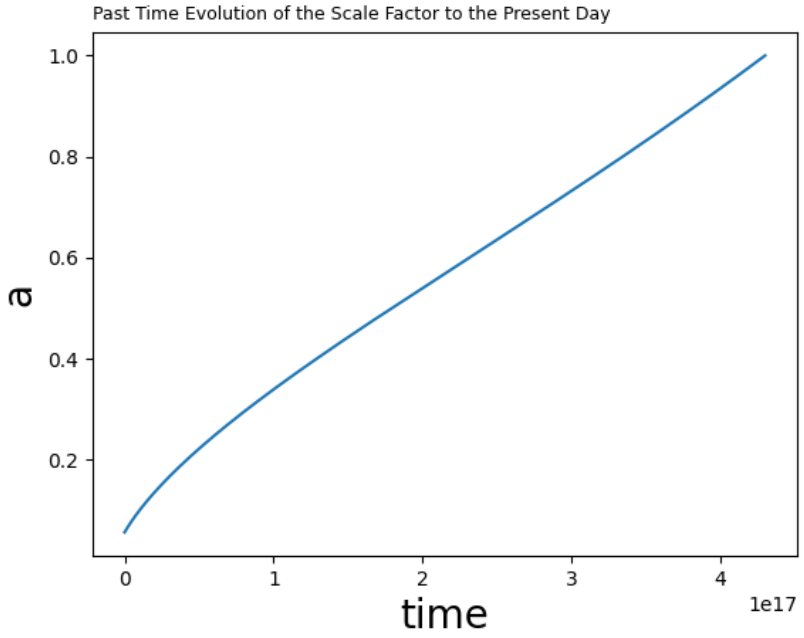


Figure 2.8: The past time evolution of scale factor for universe containing a mixed fluid. This graph depicts how the scale factor increased as a function of time, from the Big Bang singularity to present day, in the Hot Big Bang model. During the radiation dominated era, we see a slightly accelerated initial expansion that quickly stabilises into a more constant rate of expansion as the energy density of the radiation dilutes as discussed in Section 2.0.1.

Figure 2.9

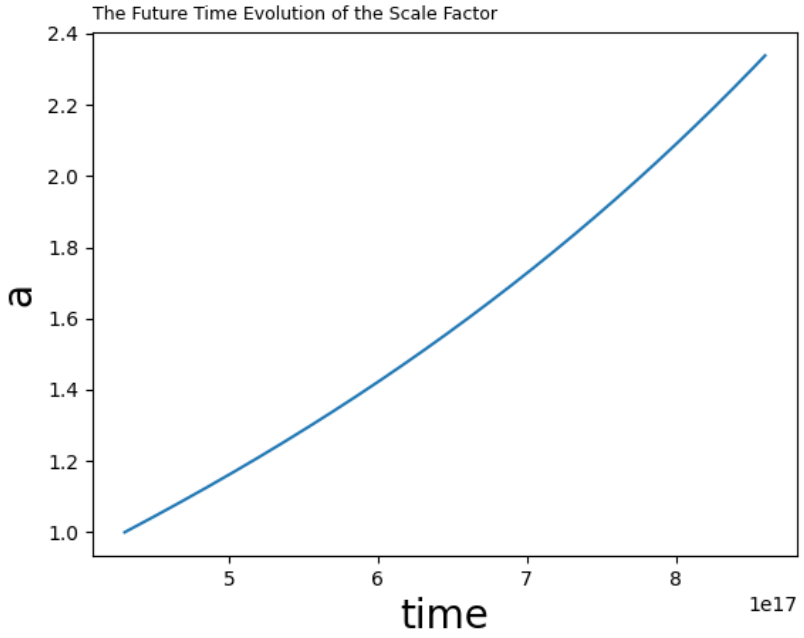


Figure 2.9: The future time evolution of the scale factor for a universe containing a mixed fluid. This graph clearly demonstrates the domination of the cosmological constant discussed in Section 2.0.1 after the energy density contribution from matter has also been diluted. This graph shows how this trend starts approximately at our current age of the Universe.

Figure 2.10

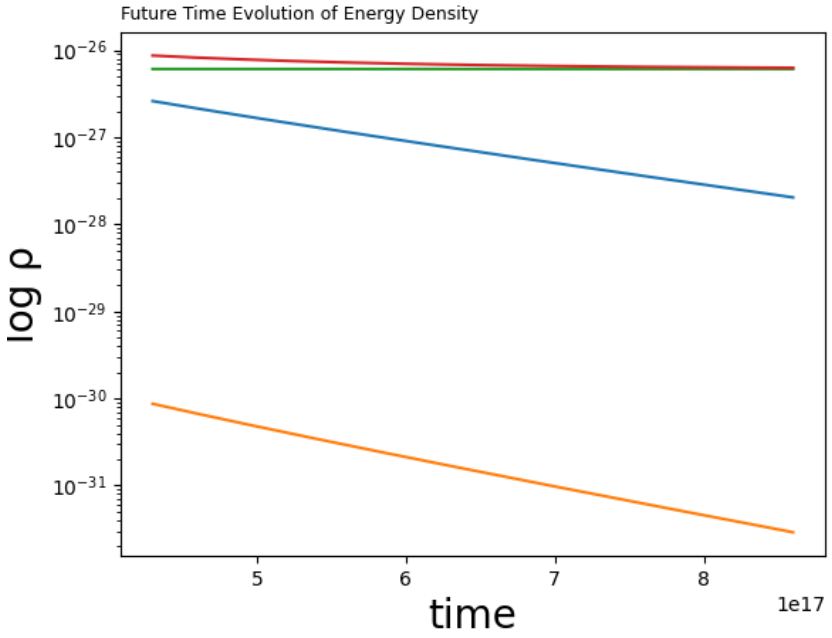


Figure 2.10: The future time evolution of energy density for a mixed fluid universe. This graph contains the contributions of matter, radiation, and the vacuum energy, as well as the total energy density. Here the use of logarithmic scaling emphasises the relative decay rates of the contributions to the energy density of different sources. Here, the steeper gradients of the energy densities corresponding to matter and radiation can be seen decaying below the constant line of the vacuum energy, despite the comsological constant orignally having a negligible effect in terms of the total contribution.)

Figure 2.11

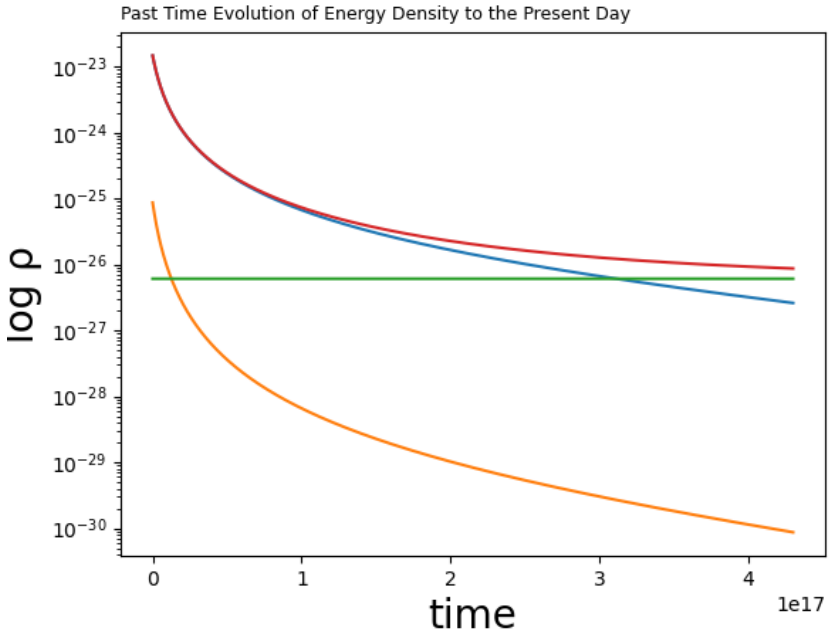


Figure 2.11: The past time evolution of energy density. This graph also depicts the contributions of matter, radiation, and the vacuum energy individually, as well as the total energy density. As we look to the past, the initial overwhelming influence of matter and radiation is even more dramatic than in Fig. 2.10.

Figure 2.12

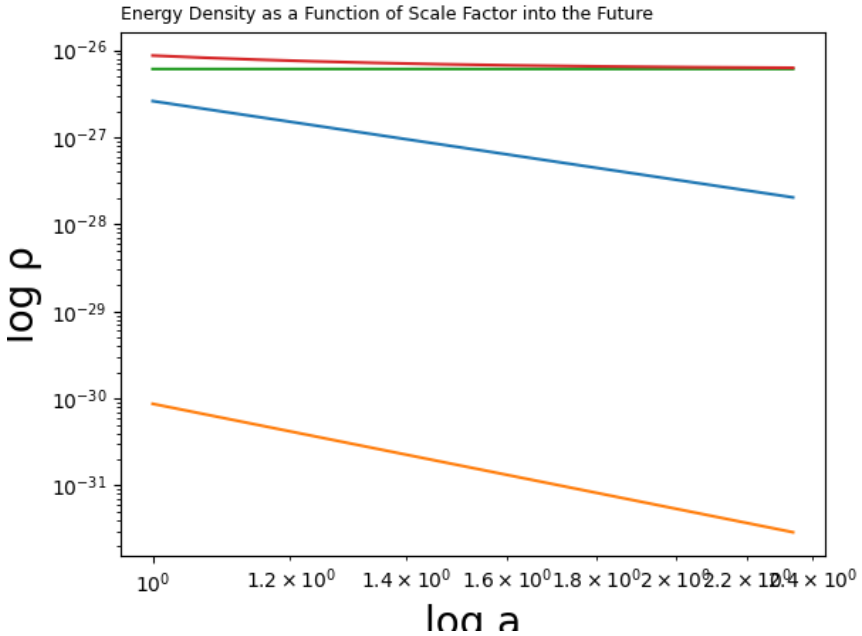


Figure 2.12: The future evolution of energy density as a function of scale factor. Note the relatively high contribution of the vacuum energy on the scale factor as we go forward in time. We will further discuss the consequences of this eventual domination in Section 2.0.3

Figure 2.13

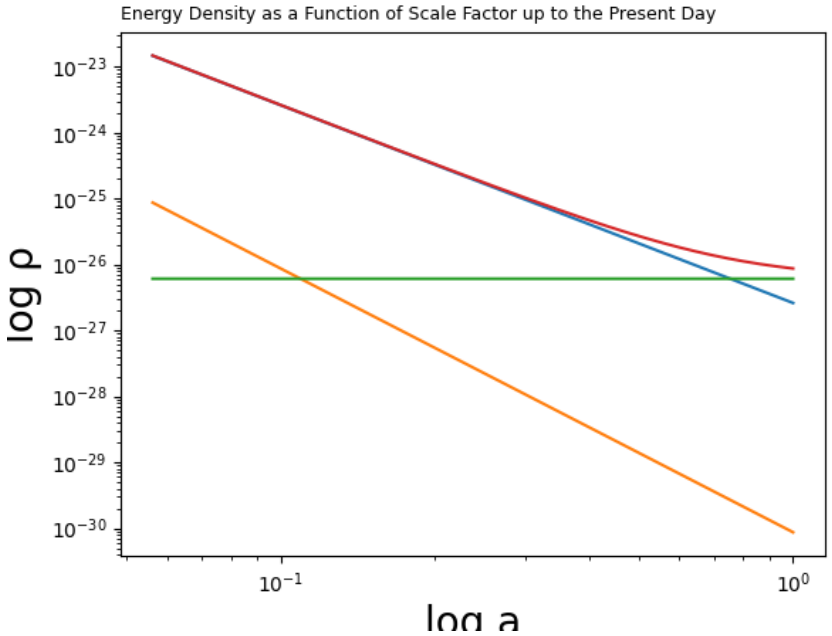


Figure 2.13: The past evolution of energy density as a function of Scale Factor. Here the early dominance of Matter and Radiation discussed in Sections 2.0.1 and 2.0.1 can be seen. It is only as we come very close to the present time, t_0, we see any significant divergence of the total energy density from the leading contribution, and it begins its approach to ρ_Λ

2.0.3 Success of the Hot Big Bang Model, and its implications for the Standard Model of Cosmology

It is hard to overstate the success of this model. Light from distant galaxies has been measured to be systematically redshifted in a manner consistent with universal expansion (Hubble [1929]). Ratios of certain elements match what is predicted by Big Bang Nucleosynthesis (Alpher et al. [1948]). And the detection of the Cosmic Microwave Background (CMB) can only be explained as a relic of the photon decoupling that occured during the recombination era (Dicke et al. [1965]).

Here we will give an overview of the Hot Big Bang model's physical implications on the current standard model of cosmology. By considering the temperatures associated with this model with other established physical theories, we learn that the Universe, at some finite point in the past, was extremely hot, dense, and small. It grew in size and cooled, which according to Quantum Field Theory

(QFT) allowed symmetry breaking, and the production the fundamental forces and particles we observe today (Griffiths [2020]). The universe's expansion during the first era was dominated by the influence of radiation. As the universe grew, perhaps on the timescale of seconds or minutes, it become cool enough for the first composite subatomic particles, such as neutrons and protons, to form. Soon after, nucleosynthesis occurred for the first time allowing nuclei to form and fuse (Tanabashi et al. [2018]). The ratio of protons and neutrons determined the relative abundance of hydrogen and helium formed as well as traces of other heavier nuclei and, from the theory of Big Bang nucleosynthesis, can be quantitatively predicted. In the present day, areas of interstellar gas unaffected by the consequent stellar nucleosynthesis have preserved these ratios of elements. The relative abundances of these elements observationally agree with the ratios predicted by Big Bang Nucleosynthesis (Loeb [2006]). The universe continued to grow and cool to the point where nuclear fusion was no longer possible, though the temperature remained high enough to ensure that all matter existed as plasma, and thus opaque to photons. The Hot Big Bang model predicts, after thousands of years, that temperatures would keep cooling to the point that electrons could then combine with the primordial nuclei to form atoms, initiating the era of recombination. This transition of matter from a plasma to gas, was followed by the newly-birthed atoms dropping to their ground state, releasing photons via photon decoupling.

Indeed, a few billion years later, a few of those same photons would be the cause of the wrongful eviction of some pigeons nesting within a six metre horn antenna, as Arno Penzias and Robert Wilson struggled in vain to locate the source of the persistent microwave hum interfering with their radio reciever (Lof [1980]). It was Penzias and Wilson's eventual realisation of the agreement between the characteristics of the radiation they detected with theoretical predictions of radiation produced from decoupling, that would convey the significance of this mysterious microwave background's source and serve as a key substantiation of this Hot Big Bang model, as well as award them the 1978 Nobel Prize for Physics (Penzias and Wilson [1965]). It would come to be known as the Cosmic Microwave Background (CMB).

The earlier domination of radiation on the expansion of the universe gives way instead to matter, as the Universe continued to expand, both diluting the number density of photons as well as red-shifting their wavelength through the expansion of space itself, as seen in case 2.0.1. Following the recombination era slight overdensity in the primordial gas clouds of hydrogen and helium initiated gravitational collapse that led to large structure formation and the eventual birth of

Population I stars (these have yet to be definitively observed). Eventually, on the scale of millions to billions of years, galaxies form, arranged in clusters and along cosmic filaments, in a foam-like structure, likely the remnants of quantum fluctuations from a time when the entire observable universe was smaller than an atomic nucleus (Liddle [2015]). The Hot Big Bang model predicts that the expansion of the universe is currently transitioning from the influence of matter to the domination of the cosmological constant, or dark energy, (clearly seen in Fig. 2.12) causing the accelerated expansion observed in the redshift of distant extragalactic supernovae.

The Hot Big Bang model not only describes the universe's growth, but also its demise. The discovery of the eternal era of accelerated expansion indicates that "Heat Death" will be the most likely future of the universe. Neighbouring galaxy clusters and eventually individual galaxies, become causally disconnected as they mutually expand out of each other's cosmological horizons. If intelligent life exists in the far future, they may have no way of knowing that other galaxies existed in the universe. Eventually the last generations of stars will deplete the material required for fusion, and long after the light of the last star is quenched, only black holes and decaying stellar remnants will remain, which too will decay after eons through hawking radiation and quantum effects (Dyson et al. [2002]). Eventually all will succumb to the Second Law of Thermodynamics - and the universe will become a structureless void for eternity as it inexorably marches towards total thermodynamic equilibrium, and entropy will reach a maximum. The Hot Big Bang model has led us away from the naive assumptions of steady-state cosmology, and provided not only detailed history of the evolution of the universe from the earliest moments, and a predictions of its prolonged and mildly disturbing demise, but most importantly a great array of verifiable and quantitative predictions, that have been verified in nearly all cases.

2.0.4 Observations that remain unexplained with the Standard Model of Cosmology

Yet despite the success of this model in explaining the Redshift-distance relation observed by Hubble of astronomical standard candles, the existence and temperature profile of the redshifted Cosmic Microwave Background Radiation (CMB), and the ratio of elements in primordial gas clouds, it appears mysterious that the CMB appears to be of uniform temperature in regions corresponding to causally disconnected regions of space-time. This homogeneity is difficult to explain, as

regions of the universe that are far apart from each other should not have had time to come into thermal equilibrium, given the age of the universe. Their light-cones, in this model, do not intersect. In fact these causally disconnected regions of the CMB not only share the same temperature, but in fact exhibit a pattern of anisotropy in temperature consistent with interaction.

Furthermore, even if the Universe began as nearly flat, any slight overdensity of matter would result in the universe becoming more and more curved as it grew in size. It would take an extremely precise and unstable initial distribution of matter to ensure the flatness we currently observe. One can show this mathematically by rewriting Eq. 1.10 and Eq. 1.14 in terms of the density parameter $\Omega \equiv \frac{\rho}{\rho_c}$. This is the ratio of some energy density with the energy density required for a flat universe ρ_c. The density parameter can be shown to evolve with the scale factor as

$$\frac{d\Omega}{d\ln a} = (1+3\omega)\Omega(\Omega-1) \tag{2.21}$$

where ω is the equation of state constant described in Eq. 2.2. If the universe is comprised of matter or radiation such as in the case of Section 2.0.1 or Section 2.0.1, then a flat universe where $\Omega = 1$ is an unstable fixed point

$$\frac{d|\Omega-1|}{d\ln a} > 0 \tag{2.22}$$

A more explicit treatment can be found in Kinney [2009].

Additionally, magnetic monopoles seem to be mathematically consistent with our current understanding of electrodynamics, given by Quantum electrodynamics. Yet they have not been observed in neither experiment nor nature, despite the absence of a clear reason to forbid their existence. Collectively, these problems are be categorised together as "fine-tuning" problems. Together they strongly indicate that the Hot Big Bang model is not entirely complete. We find that the addition of an early era of accelerated expansion in the universe's history, can explain this fine-tuning problems.

3 Inflation

In the 1970s, Zeldovich and others in the Soviet Union identified these so-called fine-tuning problems with the Hot Big Bang cosmology, and Alexei Starobinsky proposed an era of rapid exponential expansion, using quantum corrections to general relativity . One can see that the lack of monopoles is to be expected with such a theory - any formed in the initial genesis of such a universe would be flung apart so widely that one would need to be extremely fortunate to stumble across one. Furthermore, an inflationary era allows thermal contact in areas of space, that are then seperated by inflation to be causally disconnected as observed currently in the CMB. And finally, the exponential expansion of inflation smooths out any initial spacetime curvature, resulting in a euclidean universe (Kinney [2009]). The new question is now a matter of finding a physical cause for the rapid early expansion.

The physics responsible for early exponential expansion of the universe cannot be Einstein's cosmological constant Λ, as a universe dominated by vacuum energy will remain dominated by its influence for eternity, as matter ($\rho \propto \frac{1}{a^3}$) and radiation ($\rho \propto \frac{1}{a^4}$) are diluted exponentially quickly. It is impossible to reach either a radiation-dominated era or a matter-dominated era of expansion, if both of their contributions have already been blown into insignificance from the earliest epoch of the universe by Λ; and so we will never see a Hot Big Bang.

Development of Slow Roll Inflation

Alan Guth introduced these new theories with the name "inflation" in 1981 during his own attempts to grapple with the monopole problem. His attempt used false vacuums (within the framework of QFT) that decayed through bubble nucleation initiated by quantum tunnelling, but it failed to generate the required reheating when the inflation lasted long enough to actually solve fine-tuning problems that necessitated its orginal development. The answer to this was found by Linde and Albrecht-Steinhardt. Inflation could triggered by a scalar field "rolling down" a potential energy hill, instead of tunnelling out of a false vacuum state. Inflation

occurred when the field rolled slowly relative to the expansion of the universe, and ended when the hill became steeper, resulting in a reheating period. It is specifically these slow roll theories that we investigate, through similar numerical techniques utilised in the modelling of non-inflationary expansion earlier, but it is worth noting that the use of scalar fields to explain inflation is one approach of many, and can't necessarily be falsified until further observational or theoretical evidence is found to indicate a precise physical mechanism behind inflation.

Scalar Fields and the Slow Roll Conditions

In physics, a field simply assigns scalars (such as the Higgs field in QFT), vectors (such as in Maxwell's classical theory of Electromagnetism), or more generally any other kind of tensor (such as those in Eq. 1.1) to all points in some space. In a scalar field inflation theory, as the keen reader may already have anticipated, the simplest type of field is used - a scalar field. At each point in space we assign only a single directionless magnitude, that corresponds with the vacuum energy responsible for inflation. As discussed previously, the vacuum energy responsible for inflation must decay with time. We use a scalar field φ, with an equation of motion given by

$$\ddot{\varphi} + 3H\dot{\varphi} + \partial_{\varphi} V = 0 \tag{3.1}$$

where we use Newton's dot notation to denote time derivatives, and Jacobi's delta to denote the partial derivative with respect to whatever the the scalar field φ. Eq. 3.1 is not an equation of motion in the Newtonian sense, but is instead so named as it describes how the scalar value of the field at each point in space changes with time, and its rate of change. V is the potential of the scalar field, and it describes how its energy changes throughout time and space. The specific form of the potential will dictate the scalar field's behaviour and thus will have significant consequences upon how the Universe inflates. The choice of potential cannot neccessarily be specified solely from theoretical considerations, at least not as of yet. Observational data can help reject innappropriate choices, but the precise choice of potential may be somewhat arbitrary. In this report we investigate three polynomial potentials - a quartic potential in 3.0.1, a cubic potential in 3.0.1, a quadratic potential in 3.0.1 - as well as the previously alluded to potential originally derived by Starobinsky, in 3.0.1.

A homogoneous scalar field takes the form of a perfect fluid with energy density given by

$$\rho = \frac{1}{2}\dot{\varphi}^2 + V(\varphi) \tag{3.2}$$

and pressure given by

$$p = \frac{1}{2}\dot{\varphi}^2 - V(\varphi) \tag{3.3}$$

The hubble parameter, by considering Eq. 1.10, can be explicitly written in terms of the scalar field as follows

$$H^2 = \frac{8\pi}{3m_{PL}}(\dot{\varphi} + V(\varphi)) \tag{3.4}$$

Where m_{PL} is the Planck mass:

$$m_{PL} \equiv \sqrt{\frac{hc}{G}} \tag{3.5}$$

For the case of slow roll inflation, we assume that the "friction term", $3H\dot{\varphi}$, dominates Eq. 3.1. When the slow roll conditions $\dot{\varphi}^2 \ll V(\varphi)$ and $\ddot{\varphi} \ll 3H\dot{\varphi} + \partial_\varphi V$ are taken in conjunction with Eq. 3.1, we have the equation of motion of the scalar field for the slow roll approximation:

$$3H\dot{\varphi} + \partial_\varphi V = 0 \tag{3.6}$$

.

By substituting $P = \dot{\varphi}$, into Eq. 3.6 a system of coupled first order ODEs is obtained, and it can now be solved, either analytically or through numerical methods as done previously.

We cannot, through observation, constrain the initial conditions of the scalar field, it is completely unknown what energy and "velocity" it had in the earliest epochs of the universe. However, in Fig. 3.1, depicting some arbitrary potential with a single and easily accessible global minimum, one can imagine that the scalar field φ - illustrated by a ball rolling along the potential - regardless of initial velocity and value, will eventually be slowed by the "friction" of the potential, and slowly roll towards the minimum.

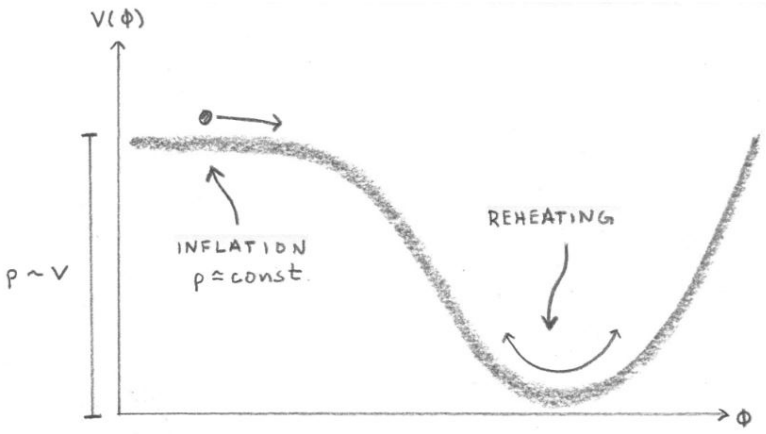


Figure 3.1: Credit to Kinney [2009] for the illuminating illustration.

When φ is "rolling" along the nearly flat section of the curve, the energy of the scalar field is non-zero and consistent, and inflation occurs. As φ slowly rolls down the potential, the previous slow roll conditions now apply. As φ reaches the bottom of the potential it oscillates about the minimum. This period is where reheating occurs. Soon, φ will have settled at the bottom, and the scalar field will have completely decayed, allowing the universe to enter into its radiation dominated era, where the Standard Model of Cosmology can take over the narrative. This property of the universe exit the period of inflation at the same value regardless of choice of initial conditions in this model, is a result of the attractor properties of the equations, and it solves the problem of ambiguity regarding the initial conditions.

Numerical and Analytic investigation of the Slow Roll Scalar Field Equations

We demonstrate this property, first by defining several equations in Python: the equations of a given potential and its derivative; the scale factor (in terms of the potential and velocity of scalar field); the Hubble parameter, and the slow roll scalar field equations of motion in Python. We further define an array of varying initial conditions for the scalar field, and the program then iterates through every set of initial conditions, calculating the value of the scalar field and its velocity at each time step by integrating over the coupled differential equations derived from Eq. 3.6. When the evolution of the scalar field for a range of different initial conditions are calculated and then plotted onto the same graph, the attractor properties of the inflation equations are visually apparent as a spiral in the scalar

field's phase space ($\dot{\varphi}$ versus φ). This corresponds to the oscillation around the minimum alluded to in the discussion of Fig. 3.1 and this property remains constant across all choices of potentials, though its exact form diverges from case to case significantly.

For each potential, we plot the potential as a function of the scalar field, the time evolution of the scalar field, the time evolution of the Hubble Parameter, and finally the phase space of the scalar field - all for a variety of different intial conditions. We also apply the slow roll conditions to the potential equation to analytically find the relation of $\dot{\varphi}$ versus φ where the slow roll approximation is valid. This manifests as the "attractor line" in the phase space and in Fig. 3.1 this corresponds to the "slow roll" of φ down the slope. We plot this relation as a dashed line in each phase space, and give the specific derivation for each potential below.

3.0.1 Scalar Inflation Field Plots

Quadratic Potential

The quadratic potential is a simple 2nd degree polynomial, given by

$$V(\varphi) = \lambda m_{PL}^2 \varphi^2 \tag{3.7}$$

Thus the derivative of the potential derivative with respect to the scalar field is given by

$$V'(\varphi) = 2\lambda m_{PL}^2 \varphi \tag{3.8}$$

Substituting this and Eq. 3.4 into Eq, 3.6 we rearrange and find the relation of $\dot{\varphi} \propto \varphi$. It is this relation, only valid during when the slow roll conditions are met, that we plot with a dashed line in Fig. 3.5.

Figure 3.2

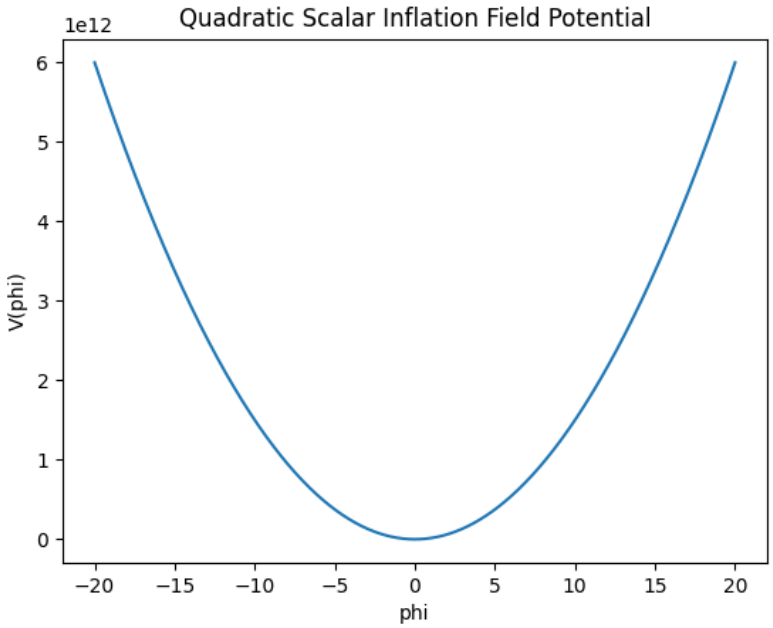


Figure 3.2: Quadratic Potential.

Perhaps the simplest of all potentials, the polynomial function of Fig. 3.2 has one well defined, and obvious global minima. If we picture the phase space produced by φ as it rolls down this potential as described in Fig. 3.1, we would expect to see a clear and simple spiral as it oscillates around the minimum.

Figure 3.3

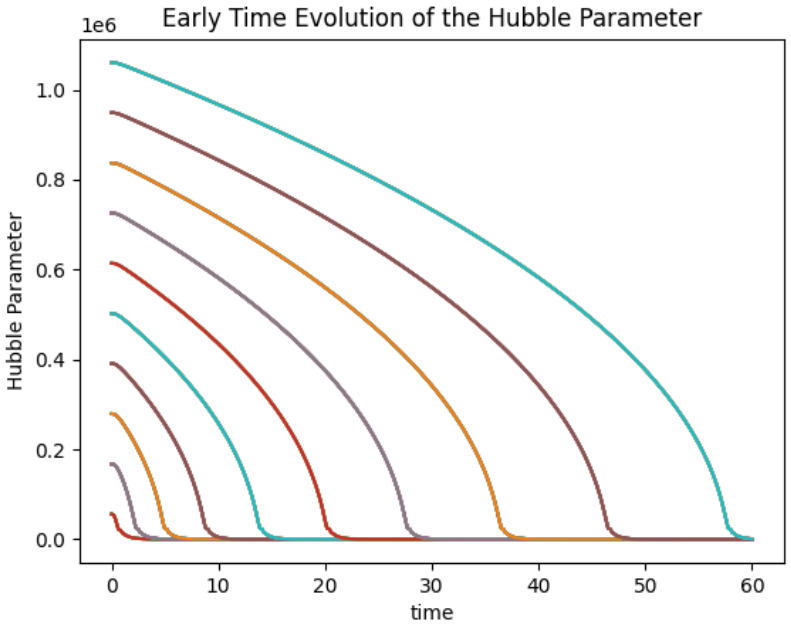


Figure 3.3: Time Evolution of the Hubble parameter with a quadratic potential.

We see in 3.3 that for a variety of initial conditions of φ, the Hubble parameter essentially follows the same general pattern of Behaviour, an accelerating approach towards a constant value of 0.

Figure 3.4

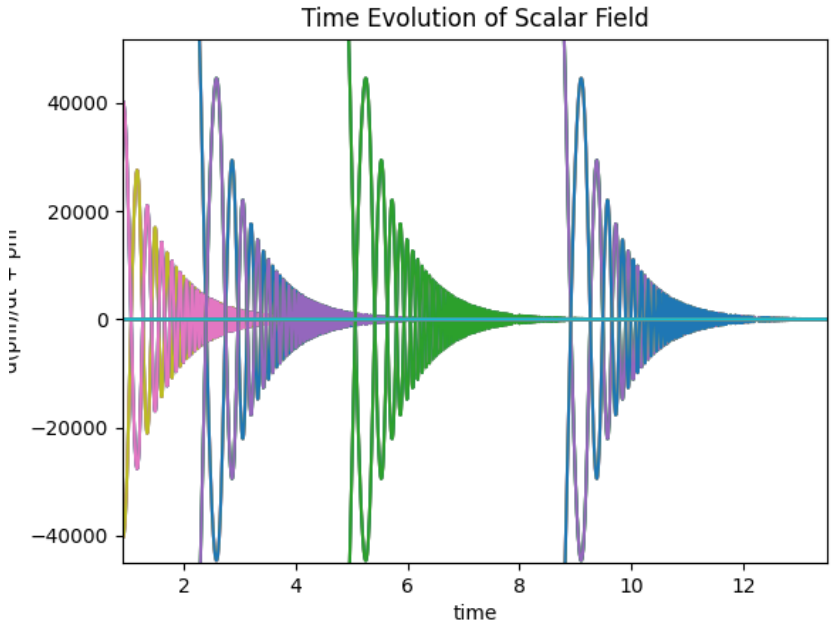


Figure 3.4: Detail of the Time Evolution of φ with a quadratic potential.

Fig. 3.4 produces another perspective of the osscilations of the scalar field around the minimum. Here we see the most significant impact of the initial conditions, on this scale the time it takes to decay away completely varies significantly.

Figure 3.5

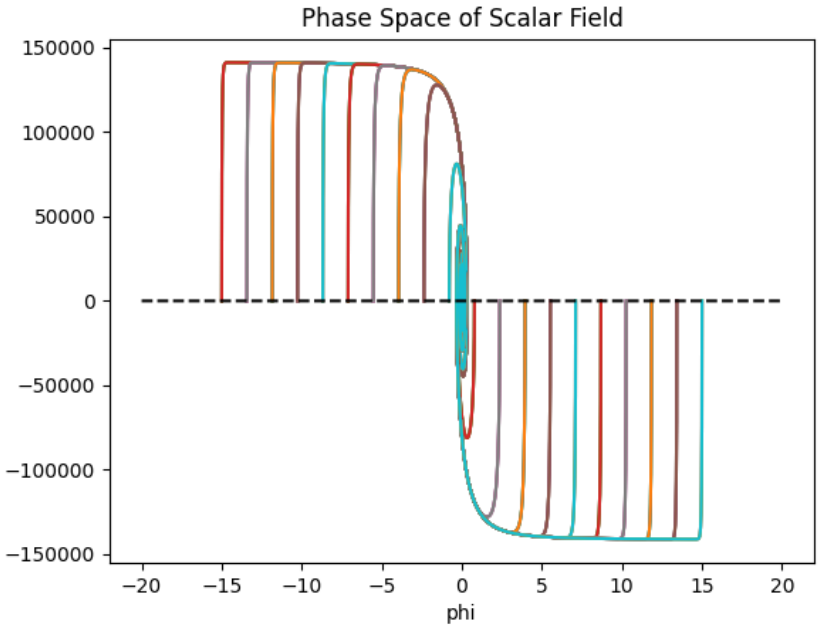


Figure 3.5: Phase space of φ with a quadratic potential.

In Fig. 3.5, as expected from the simplicity of Fig. 3.2, the phase space is simple and well behaved.

Figure 3.6

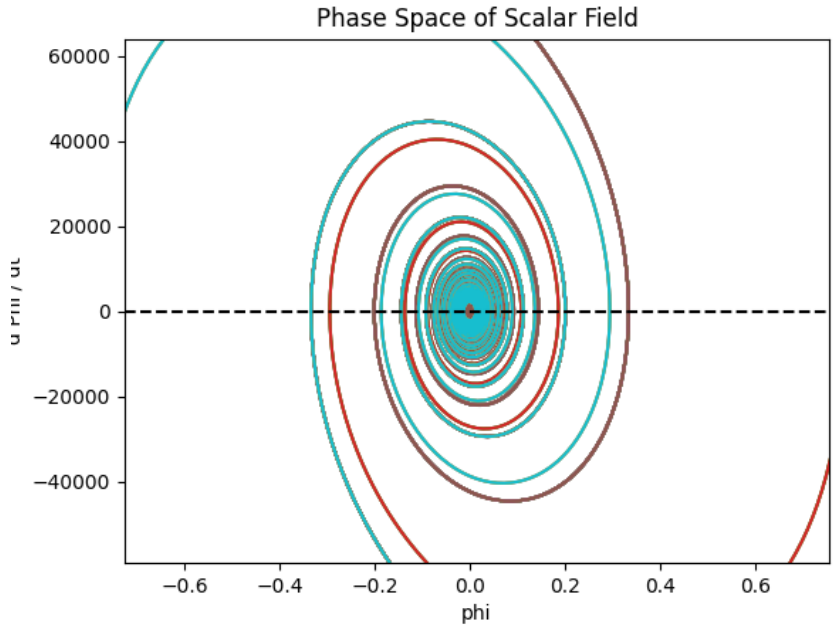


Figure 3.6: Zoomed-in Phase space of φ with a quadratic potential. Here we clearly see the symmetric approach to the minimum value as the scalar field decays, producing an era of reheating as the universe exits in the inflationary era.

Cubic Potential

The cubic potential is a simple 3nd degree polynomial, given by

$$V(\varphi) = \lambda m_{PL} \varphi^3 \tag{3.9}$$

The derivative of the potential derivative with respect to the scalar field is given by

$$V'(\varphi) = 3\lambda m_{PL} \varphi^2 \tag{3.10}$$

Again we substitute this and Eq. 3.4 into Eq, 3.6 to rearrange and find the relation of $\dot{\varphi} \propto \varphi$ during the slow roll. We then plot this curve with a dashed line in Fig. 3.10.

Figure 3.7

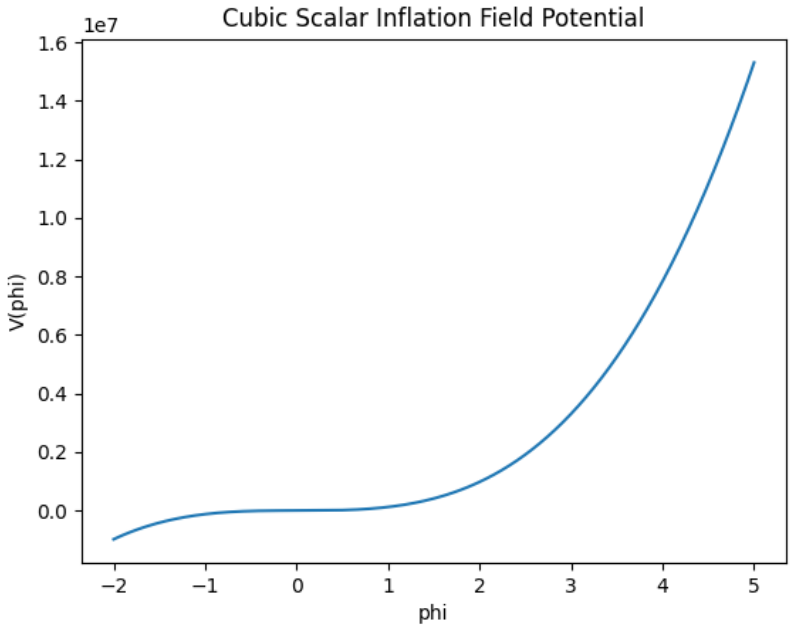


Figure 3.7: Cubic potential.

If we consider the behaviour of a scalar field in this potential, we recognise the limitations this choice of potential has on the allowed initial conditions. We can see in 3.7 while scalar field can approach the minimum from the right, an approach from the left would result on the scalar field failing to converge to a energy of zero. Thus it will not decay away, instead becoming an attractive field due to its negative energy. This behaviour is a result of the fact that the cubic polynomial used for the potential has only a local minimum at $\varphi = 0$.

Figure 3.8

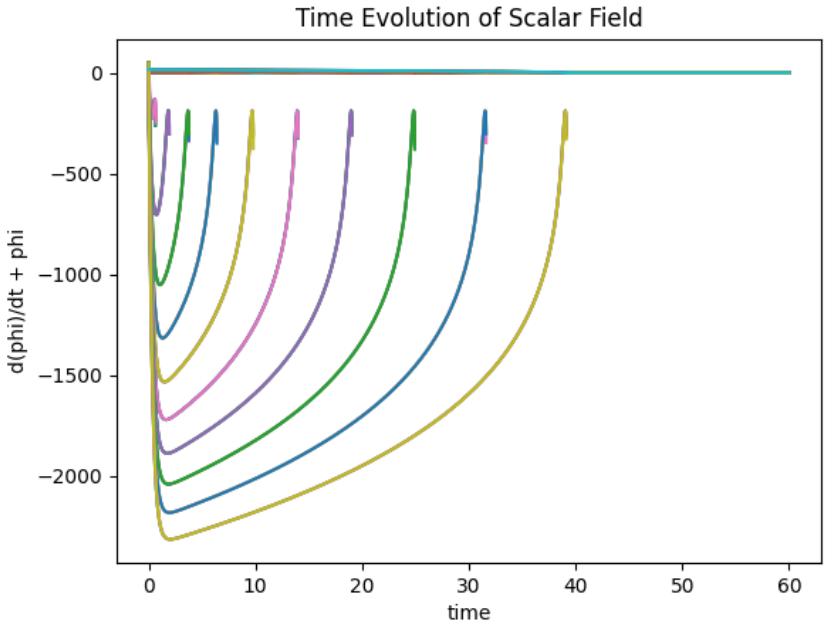


Figure 3.8: Time Evolution of φ with a cubic potential

Figure 3.9

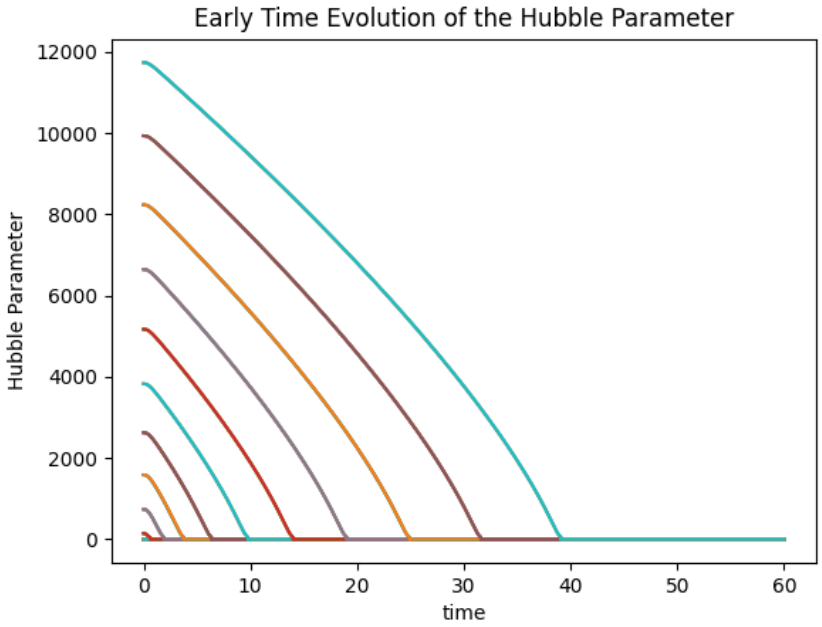


Figure 3.9: The time evolution of the Hubble Parameter with a cubic potential. For the range of initial conditions that do correspond with a slow roll of the scalar field to zero, we see appropriate behaviour of the Hubble Parameter, where after a period off inflation, approaches a low and apparently constant value.

Figure 3.10

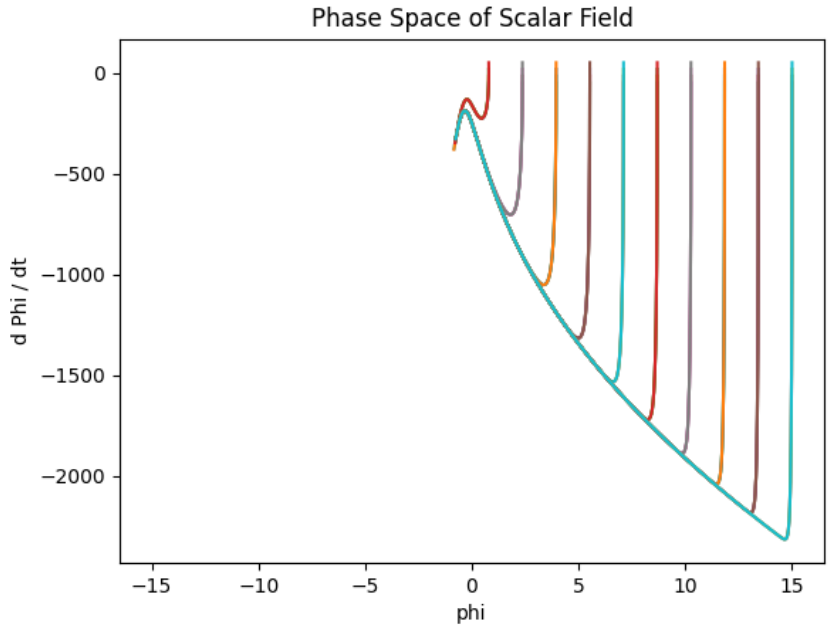


Figure 3.10: Phase space of φ with a cubic potential. The consequence of the asymmetry of the potential can clearly be seen. In fact of all the potentials investigated, this is the only one that fails to produce a spiral pattern corresponding with the approach to the minimum. This may be due to the very slight left most "boundary" of the potential. Whereas in Fig. 3.2 a scalar field with enough velocity to overshoot the minimum will roll back, we see in Fig. 3.7 that for this choice of cubic potential any scalar field with sufficient velocity to overshoot the local minimum will not encounter sufficient resistance to oscillate back.

Quartic Potential

The quartic potential is a simple 4nd degree polynomial, given by

$$V(\varphi) = \lambda\varphi^4 \tag{3.11}$$

The derivative of the potential derivative with respect to the scalar field is given by

$$V'(\varphi) = 4\lambda\varphi^3 \tag{3.12}$$

When we substitute $V'(\varphi)$ and $H(\varphi)$ into Eq, 3.6 and rearrange we find the relation of $\dot{\varphi} \propto \varphi$ and then plot this curve with a dashed line in Fig. 3.14.

Figure 3.11

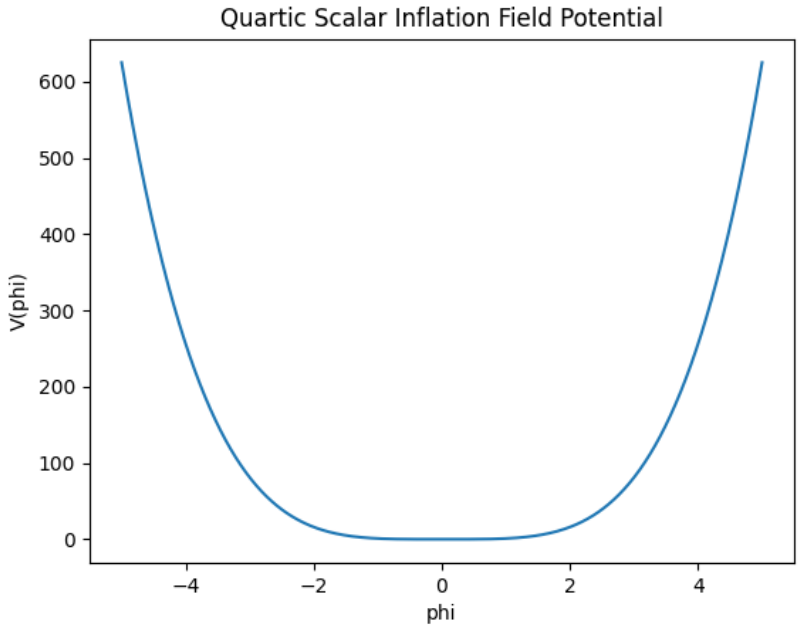


Figure 3.11: Quartic potential. This potential combines the slight inflection of the cubic potential with the symmetry of the quadratic potential. We would expect to see a scalar field that does oscillate around the minimum but in a more complex way than seen in Fig. 3.2.

Figure 3.12

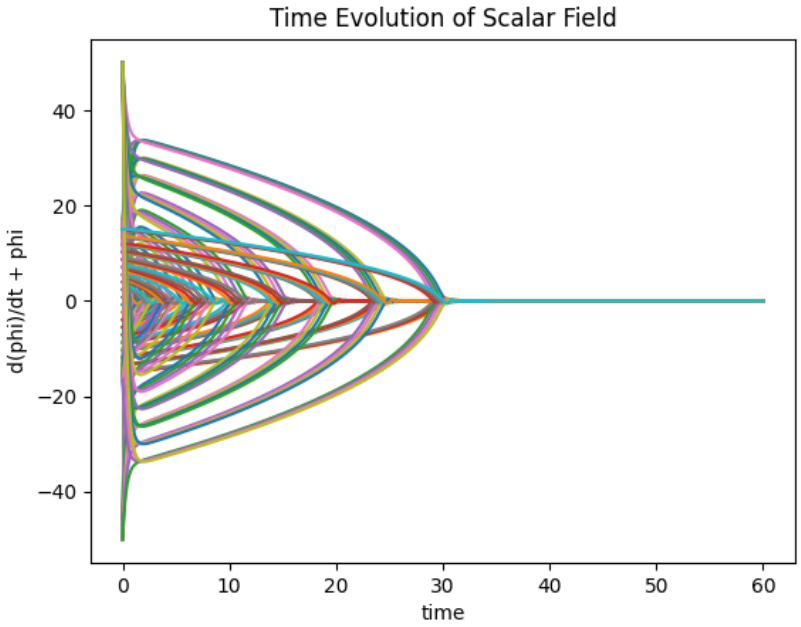


Figure 3.12: Time Evolution of φ with a quartic potential

Fig. 3.12 shows how the scalar field decays with time, for a variety of different starting conditions. Bearing in mind the relation between the magnitude of the scalar field $|\varphi|$, and the Hubble parameter H, we can anticipate the congruency between this figure and Fig. 3.13.

Figure 3.13

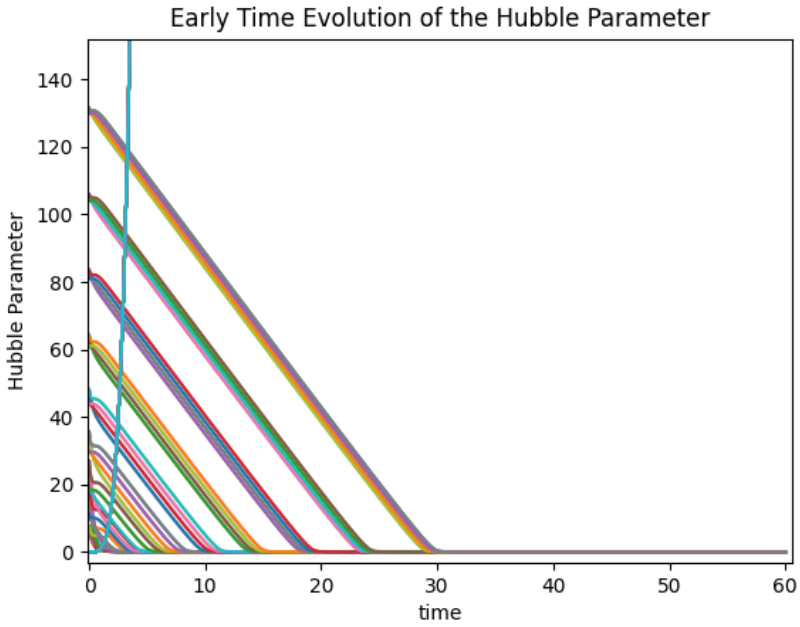


Figure 3.13: The time evolution of the Hubble Parameter with a quartic potential

The constant approach of H to 0 in Figure 3.13 diverges from Figure 3.3, whose decay seemed dependant on the inital conditions of H.

Figure 3.14

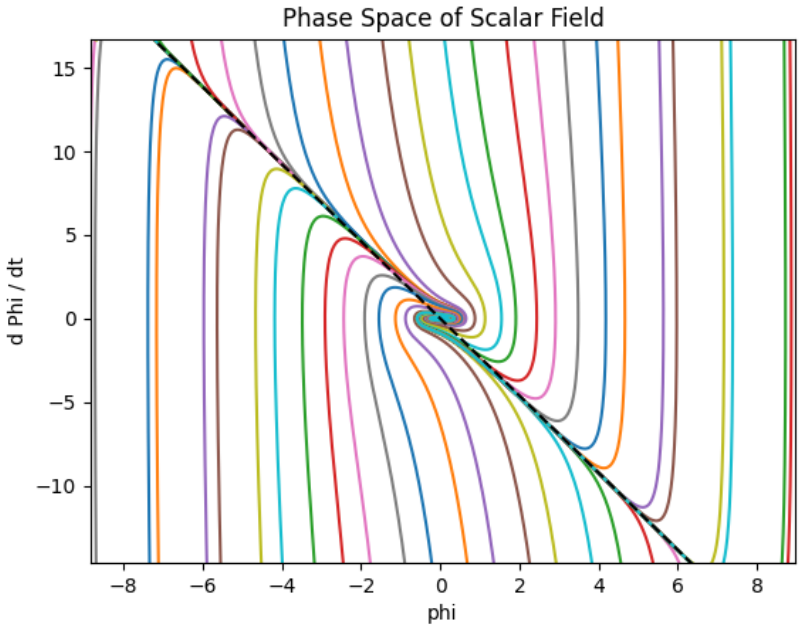


Figure 3.14: Phase space of φ with a quartic potential

Figure 3.15

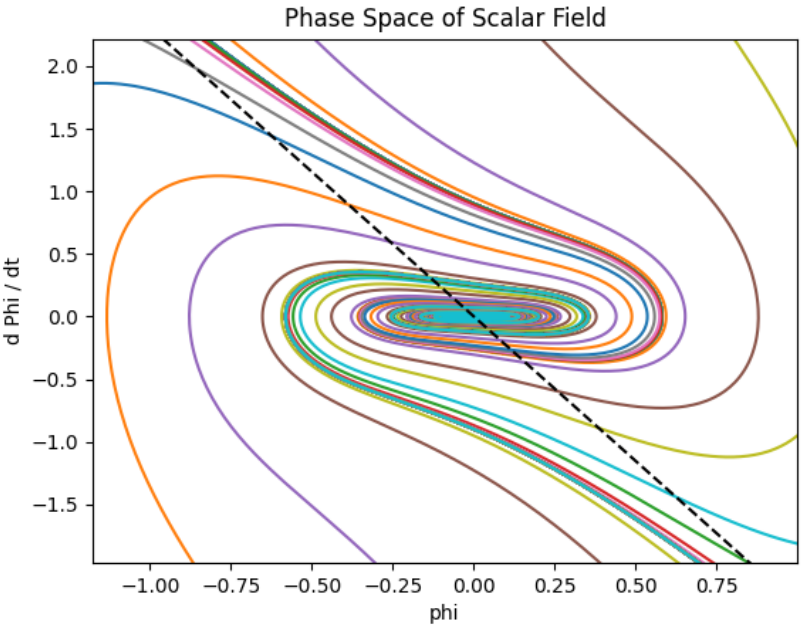


Figure 3.15: Zoomed in Phase space of φ with a quartic potential

Starobinsky Potential

The Starobinsky potential is the most complex investigated in this report. It is given by

$$V(\phi) = \lambda^4 \left[1 - e^{-\sqrt{\frac{2}{3}}\frac{\phi}{M_{\rm pl}}} \right]^2 \tag{3.13}$$

It is not a suprise then that it its derivative is also the most complex

$$V'(\phi) = \frac{2^3}{2} \cdot \frac{l^4}{\sqrt{3} \cdot M_{\rm pl}} \cdot \left(e^{\sqrt{\frac{2}{3}}\frac{\phi}{M_{\rm pl}}} - 1 \right) \cdot e^{-\frac{3}{\sqrt{3}}\phi} \tag{3.14}$$

Figure 3.16

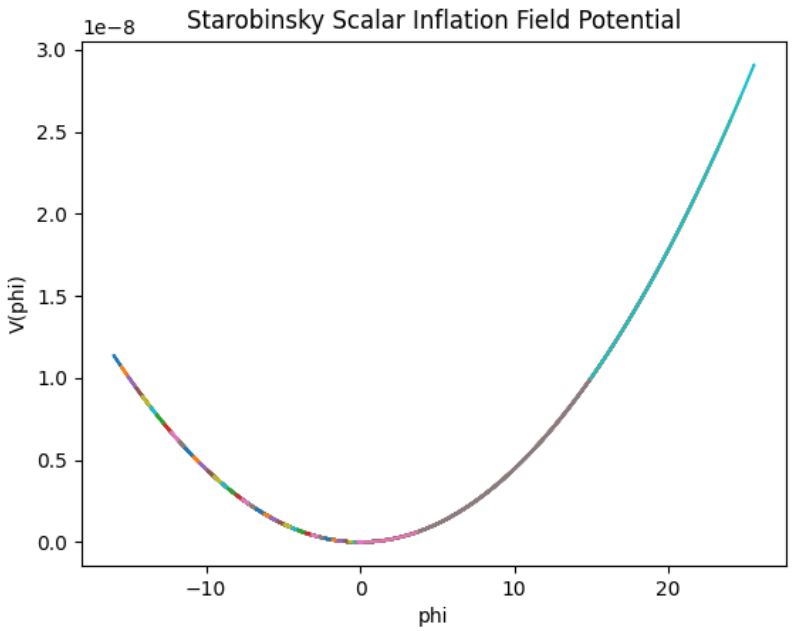


Figure 3.16: Starobinsky potential

It is suprising that the indimidating equation given in Eq. 3.14 corresponds to Figure 3.16. Locally, around the minimum, this potential could be approximated either by the quadratic or quartic potentials.

Figure 3.17

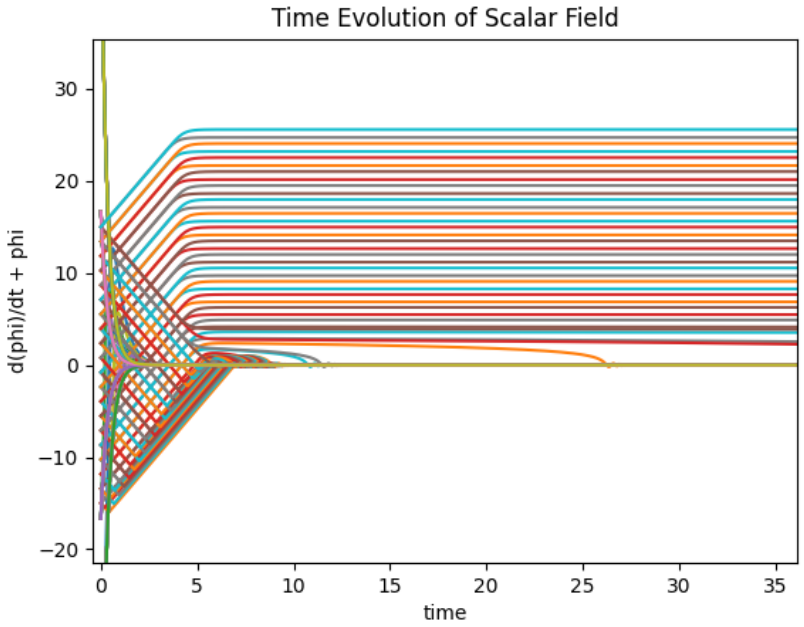


Figure 3.17: Time Evolution of φ with a Starobinsky potential

In Figure 3.17, we begin to see the increased structure of this potential. The time evolution of the scalar field is asymmetric, and there appears to be a high order of complexity between the initial condition of the scalar field and its time evolution.

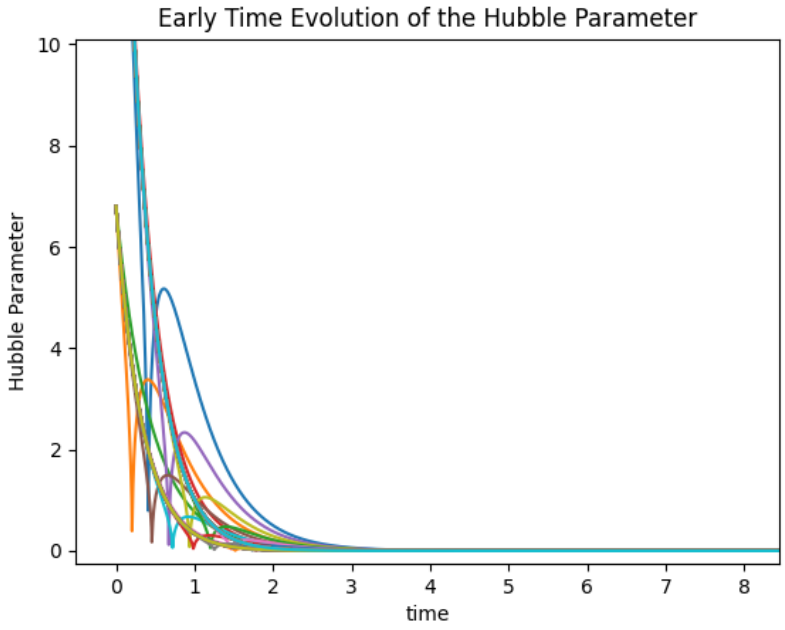


Figure 3.18

Figure 3.18: The time evolution of the Hubble Parameter with a Starobinsky potential

Figure 3.18 interestingly shows a far higher degree of correspondance between the choice of initial conditions and mode of decay. We can see the "bounce" of the Hubble parameter as it evolves forward in time, with a period proportional to its inital magnitude.

Figure 3.19

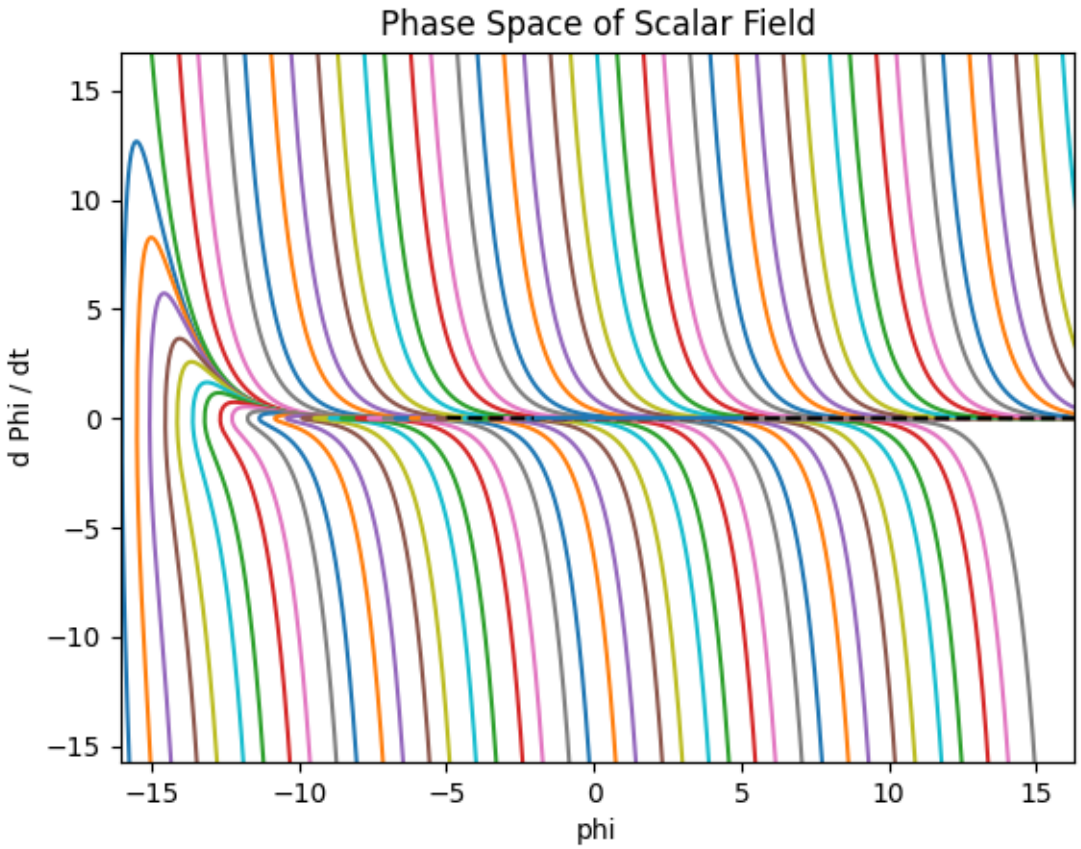


Figure 3.19: Phase space of φ with the Starobinsky potential

The scaling of the phase space obscures the attractor behaviour of the Starobinsky potential in Figure 3.19. The congruency between it and the even power law potentials, as displayed in Figure 3.16, is not clear here.

Figure 3.20

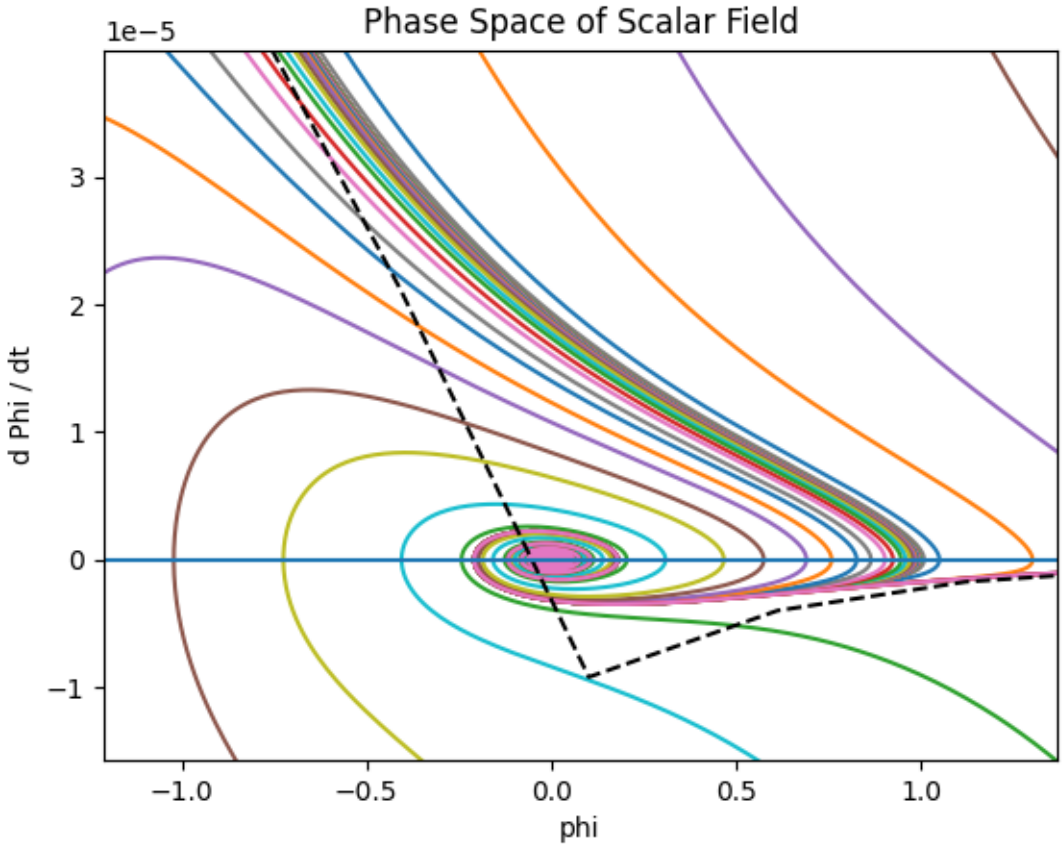


Figure 3.20: Zoomed in Phase space of φ with a Starobinsky potential

By reducing the scale of the φ axis relative to the $dot\varphi$ axis, the attractor becomes dramatically identifiable. It is reminicent of the even power law potentials as predicted by the potential, but the additional structure corresponding to the increased complexity of it's potential is also visible.

3.0.2 Parameterising Inflation with e-folds, and the Slow Roll Parameters

During inflation, the universe expands quasi-exponentially (Kinney [2009]):

$$a(t) \propto \exp\left(\int H dt\right) \equiv e^{-N} \tag{3.15}$$

Here we have defined a new parameter N, which measure's the exponential expansion of the universe. Specifically, N counts the number of e-folds for a given

expansion, where an e-folding is simply the time interval where an exponentially growing quantity increases by a factor of Euler's number e. In this case the exponentially growing quantity of interest is H, Hubble's parameter. Conventionally N is denoted with the following sign convention

$$dN \equiv -Hdt \tag{3.16}$$

or alternatively

$$N = -\int Hdt \tag{3.17}$$

such that as we travel into the past, N increases.
Here we introduce the parameter ϵ defined as

$$\epsilon \equiv \frac{3}{2}\left(\frac{p}{\rho}+1\right) \tag{3.18}$$

and secondly, the dimensionless parameter η defined as

$$\eta \equiv -\frac{\ddot{\varphi}}{H\dot{\varphi}} = \epsilon + \frac{1}{2\epsilon}\frac{d\epsilon}{dN} \tag{3.19}$$

Both parameters are related to the scalar field potential as

$$\epsilon = \frac{m_{PL}^2}{2}\left(\frac{V'(\varphi)}{V(\varphi)}\right)^2 \tag{3.20}$$

and

$$\eta = m_{PL}^2\frac{V''(\varphi)}{V(\varphi)} \tag{3.21}$$

As long as $\epsilon < 1$, inflation will take place, but if additionally both parameters are small and the potential is flat ($\epsilon, \eta \ll 1$) then the slow roll approximation is valid (Alabidi and Lidsey [2008]). Due to this fact, ϵ and η are known as the slow roll parameters, and they have been constrained by observation.

In the 2018 paper and Y. Akrami et al. [2020], spectral analysis of anistropies in the CMB were analysed to find the relative amplitude of gravitational waves to density perturbations generated during inflation.

$$r \equiv \frac{P_T^2}{P_S^2} \tag{3.22}$$

This is known as the tensor-to-scalar ratio, is directly related to the slow roll pa-

rameters. From this relation, the 2018 paper found that the slow roll parameters corresponding to power law potentials - such as those investigated in 3.0.1, 3.0.1, and 3.0.1 - were heavily disfavoured in contrast to the Starobinsky potential seen in 3.0.1. This is perhaps to be expected, due the Starobinsky potential's origin from theoretical considerations of both General Relativity and quantum fluctuations. The power law potentials appear to be more of a "toy model" in nature, useful for exploring simple behaviour but not physically applicable to realistic situations. Furthermore, the 2018 paper claimed to see no evidence of suggesting dynamics beyond a slow roll scalar field.

We suggest future study in the issues of CMB anistropy. Direct power spectrum analysis of fluctuations in the CMB and its implications on the tensor-to-scalar ratio should be investigated perhaps by utilizing Monte Carlo techniques and wavelet analysis to model the quantum fluctuations that later produced the thermal anistropies in the CMB. It would be illuminating to broadly investigate how altering characterstics of the randomly generated model affects the resultant tensor-to-scalar ratio, and thus the slow roll parameters that would describe the inflation of such a universe.

Some potentials investigated in the 2018 paper were initially derived from recently developed, and as of yet unverified, concepts stemming from String theory and Quantum Loop Gravity. Such theories attempt both to agree with succesful results from the Standard Model of Particle Physics and Quantum Field Theory, and the General Theory of Relativity, two branches of physics widely succesful and mutually irreconcilable. It is hoped that a theory that is able to unify the two may additionally explain current gaps in contemporary physics, dark energy and the precise mechanism behind inflation being one of them. In this case, any such speculative theory should predict an inflationary era that conforms with the constraints set by the 2018 paper. These results may indeed be useful in working backwards from observation, providing some guidance for theoretical developments. New physics nearly always comes from unexpected places, the discovery of Entropy came from the study of steam engines, string theory came from scattering amplitudes, and AdS/CFT correspondance had its roots in black hole thermodynamics (Susskind [1995])(Stowe [2013])(Zwiebach [2017]). Perhaps in grappling with questions surrounding slow roll scalar fields, and more generally inflation, the ideas and frameworks that emerge may prove fruitful.

Bibliography

The nobel prize in physics 2011. *NobelPrize.org*, Oct 2011. URL https://www.nobelprize.org/uploads/2018/06/press-14.pdf.

L. Alabidi and J. E. Lidsey. Single-field inflation after the WMAP five-year data. *Physical Review D*, 78(10), nov 2008. doi: 10.1103/physrevd.78.103519. URL https://doi.org/10.1103%2Fphysrevd.78.103519.

R. A. Alpher, H. Bethe, and G. Gamow. The origin of chemical elements. *Phys. Rev.*, 73:803–804, Apr 1948. doi: 10.1103/PhysRev.73.803. URL https://link.aps.org/doi/10.1103/PhysRev.73.803.

and N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Baccigalupi, M. Ballardini, A. J. Banday, R. B. Barreiro, N. Bartolo, S. Basak, R. Battye, K. Benabed, J.-P. Bernard, M. Bersanelli, P. Bielewicz, J. J. Bock, J. R. Bond, J. Borrill, F. R. Bouchet, F. Boulanger, M. Bucher, C. Burigana, R. C. Butler, E. Calabrese, J.-F. Cardoso, J. Carron, A. Challinor, H. C. Chiang, J. Chluba, L. P. L. Colombo, C. Combet, D. Contreras, B. P. Crill, F. Cuttaia, P. de Bernardis, G. de Zotti, J. Delabrouille, J.-M. Delouis, E. D. Valentino, J. M. Diego, O. Doré , M. Douspis, A. Ducout, X. Dupac, S. Dusini, G. Efstathiou, F. Elsner, T. A. Enßlin, H. K. Eriksen, Y. Fantaye, M. Farhang, J. Fergusson, R. Fernandez-Cobos, F. Finelli, F. Forastieri, M. Frailis, A. A. Fraisse, E. Franceschi, A. Frolov, S. Galeotta, S. Galli, K. Ganga, R. T. Génova-Santos, M. Gerbino, T. Ghosh, J. González-Nuevo, K. M. Górski, S. Gratton, A. Gruppuso, J. E. Gudmundsson, J. Hamann, W. Handley, F. K. Hansen, D. Herranz, S. R. Hildebrandt, E. Hivon, Z. Huang, A. H. Jaffe, W. C. Jones, A. Karakci, E. Keihänen, R. Keskitalo, K. Kiiveri, J. Kim, T. S. Kisner, L. Knox, N. Krachmalnicoff, M. Kunz, H. Kurki-Suonio, G. Lagache, J.-M. Lamarre, A. Lasenby, M. Lattanzi, C. R. Lawrence, M. L. Jeune, P. Lemos, J. Lesgourgues, F. Levrier, A. Lewis, M. Liguori, P. B. Lilje, M. Lilley, V. Lindholm, M. López-Caniego, P. M. Lubin, Y.-Z. Ma, J. F. Macías-Pérez, G. Maggio, D. Maino, N. Mandolesi, A. Mangilli, A. Marcos-Caballero, M. Maris, P. G. Martin, M. Martinelli, E. Martínez-González, S. Matarrese, N. Mauri, J. D. McEwen, P. R. Meinhold, A. Melchiorri, A. Mennella, M. Migliaccio, M. Millea, S. Mitra, M.-A. Miville-Deschênes, D. Molinari, L. Montier, G. Morgante, A. Moss, P. Natoli, H. U. Nørgaard-Nielsen, L. Pagano, D. Paoletti, B. Partridge, G. Patanchon, H. V. Peiris, F. Perrotta, V. Pettorino, F. Piacentini, L. Polastri, G. Polenta, J.-L. Puget, J. P. Rachen, M. Reinecke, M. Remazeilles, A. Renzi, G. Rocha, C. Rosset, G. Roudier, J. A. Rubiño-Martín, B. Ruiz-Granados, L. Salvati, M. Sandri, M. Savelainen, D. Scott, E. P. S. Shellard, C. Sirignano, G. Sirri, L. D. Spencer, R. Sunyaev, A.-S. Suur-Uski, J. A. Tauber, D. Tavagnacco, M. Tenti,

L. Toffolatti, M. Tomasi, T. Trombetti, L. Valenziano, J. Valiviita, B. V. Tent, L. Vibert, P. Vielva, F. Villa, N. Vittorio, B. D. Wandelt, I. K. Wehus, M. White, S. D. M. White, A. Zacchei, and A. Zonca. iplanck/i2018 results. *Astronomy &amp Astrophysics*, 641:A6, sep 2020. doi: 10.1051/0004-6361/201833910. URL https://doi.org/10.1051%2F0004-6361%2F201833910.

and Y. Akrami, F. Arroja, M. Ashdown, J. Aumont, C. Baccigalupi, M. Ballardini, A. J. Banday, R. B. Barreiro, N. Bartolo, S. Basak, K. Benabed, J.-P. Bernard, M. Bersanelli, P. Bielewicz, J. J. Bock, J. R. Bond, J. Borrill, F. R. Bouchet, F. Boulanger, M. Bucher, C. Burigana, R. C. Butler, E. Calabrese, J.-F. Cardoso, J. Carron, A. Challinor, H. C. Chiang, L. P. L. Colombo, C. Combet, D. Contreras, B. P. Crill, F. Cuttaia, P. de Bernardis, G. de Zotti, J. Delabrouille, J.-M. Delouis, E. D. Valentino, J. M. Diego, S. Donzelli, O. Doré , M. Douspis, A. Ducout, X. Dupac, S. Dusini, G. Efstathiou, F. Elsner, T. A. Enßlin, H. K. Eriksen, Y. Fantaye, J. Fergusson, R. Fernandez-Cobos, F. Finelli, F. Forastieri, M. Frailis, E. Franceschi, A. Frolov, S. Galeotta, S. Galli, K. Ganga, C. Gauthier, R. T. Génova-Santos, M. Gerbino, T. Ghosh, J. González-Nuevo, K. M. Górski, S. Gratton, A. Gruppuso, J. E. Gudmundsson, J. Hamann, W. Handley, F. K. Hansen, D. Herranz, E. Hivon, D. C. Hooper, Z. Huang, A. H. Jaffe, W. C. Jones, E. Keihänen, R. Keskitalo, K. Kiiveri, J. Kim, T. S. Kisner, N. Krachmalnicoff, M. Kunz, H. Kurki-Suonio, G. Lagache, J.-M. Lamarre, A. Lasenby, M. Lattanzi, C. R. Lawrence, M. L. Jeune, J. Lesgourgues, F. Levrier, A. Lewis, M. Liguori, P. B. Lilje, V. Lindholm, M. López-Caniego, P. M. Lubin, Y.-Z. Ma, J. F. Macías-Pérez, G. Maggio, D. Maino, N. Mandolesi, A. Mangilli, A. Marcos-Caballero, M. Maris, P. G. Martin, E. Martínez-González, S. Matarrese, N. Mauri, J. D. McEwen, P. D. Meerburg, P. R. Meinhold, A. Melchiorri, A. Mennella, M. Migliaccio, S. Mitra, M.-A. Miville-Deschênes, D. Molinari, A. Moneti, L. Montier, G. Morgante, A. Moss, M. Münchmeyer, P. Natoli, H. U. Nørgaard-Nielsen, L. Pagano, D. Paoletti, B. Partridge, G. Patanchon, H. V. Peiris, F. Perrotta, V. Pettorino, F. Piacentini, L. Polastri, G. Polenta, J.-L. Puget, J. P. Rachen, M. Reinecke, M. Remazeilles, A. Renzi, G. Rocha, C. Rosset, G. Roudier, J. A. Rubiño-Martín, B. Ruiz-Granados, L. Salvati, M. Sandri, M. Savelainen, D. Scott, E. P. S. Shellard, M. Shiraishi, C. Sirignano, G. Sirri, L. D. Spencer, R. Sunyaev, A.-S. Suur-Uski, J. A. Tauber, D. Tavagnacco, M. Tenti, L. Toffolatti, M. Tomasi, T. Trombetti, J. Valiviita, B. V. Tent, P. Vielva, F. Villa, N. Vittorio, B. D. Wandelt, I. K. Wehus, S. D. M. White, A. Zacchei, J. P. Zibin, and A. Zonca. iplanck/i2018 results. *Astronomy &amp Astrophysics*, 641:A10, sep 2020. doi: 10.1051/0004-6361/201833887. URL https://doi.org/10.1051%2F0004-6361%2F201833887.

A. G. Debus. *Man and nature in the Renaissance*. Cambridge University Press, 2004.

R. H. Dicke, P. J. E. Peebles, P. G. Roll, and D. T. Wilkinson. Cosmic Black-Body Radiation. *ApJ*, 142:414–419, July 1965. doi: 10.1086/148306.

R. DiSalle. *Understanding space-time: The philosophical development of physics from Newton to einstein*. Cambridge University Press, 2008.

L. Dyson, M. Kleban, and L. Susskind. Disturbing implications of a cosmological constant. *Journal of High Energy Physics*, 2002(10):011–011, 2002. doi: 10.1088/1126-6708/2002/10/011.

A. Einstein. English translation of "the foundation of the general theory of relativity". *The Road to Relativity*, page 183–226, 1920. doi: 10.2307/j.ctt1vwmgv8.14.

G. Goldhaber and D. B. Cline. The acceleration of the expansion of the universe: A brief early history of the supernova cosmology project (SCP). In *AIP Conference Proceedings*. AIP, 2009. doi: 10.1063/1.3232196. URL https://doi.org/10.1063%2F1.3232196.

D. J. Griffiths. *Introduction to elementary particles*. Wiley-VCH, 2020.

E. Hubble. A Relation between Distance and Radial Velocity among Extra-Galactic Nebulae. *Proceedings of the National Academy of Science*, 15(3):168–173, Mar. 1929. doi: 10.1073/pnas.15.3.168.

W. H. Kinney. Tasi lectures on inflation. *arXiv preprint arXiv:0902.1529*, 2009.

A. Liddle. *An introduction to modern cosmology*. John Wiley amp; Sons Inc, 2015.

A. Loeb. The dark ages of the universe. *Scientific American*, 295(5):46–53, 2006. doi: 10.1038/scientificamerican1106-46.

C. Lof. A scientist looks at science–an interview with arno penzias. *IEEE Communications Magazine*, 18(4):8–13, 1980. doi: 10.1109/mcom.1980.1090259.

N. D. Mermin. Is the moon there when nobody looks? reality and the quantum theory. *Physics Today*, 38(4):38–47, 1985. doi: 10.1063/1.880968.

C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. Freeman, 1995.

C. O'Raifeartaigh, M. O'Keeffe, W. Nahm, and S. Mitton. Einstein's 1917 static model of the universe: A centennial review. *The European Physical Journal H*, 42 (3):431–474, 2017. doi: 10.1140/epjh/e2017-80002-5.

L. Pasquini, P. Bonifacio, S. Randich, D. Galli, and R. G. Gratton. Beryllium in turnoff stars of ngcnbsp;6397: Early galaxy spallation, cosmochronology and cluster formation. *Astronomy amp; Astrophysics*, 426(2):651–657, 2004. doi: 10.1051/0004-6361:20041254.

P. J. E. Peebles and B. Ratra. The cosmological constant and dark energy. *Rev. Mod. Phys.*, 75:559–606, Apr 2003. doi: 10.1103/RevModPhys.75.559. URL https://link.aps.org/doi/10.1103/RevModPhys.75.559.

A. A. Penzias and R. W. Wilson. A Measurement of Excess Antenna Temperature at 4080 Mc/s. *ApJ*, 142:419–421, July 1965. doi: 10.1086/148307.

A. G. Riess, S. Casertano, W. Yuan, L. M. Macri, and D. Scolnic. Large magellanic cloud cepheid standards provide a 1% foundation for the determination of the hubble constant and stronger evidence for physics beyond {$upLambda$aCDM. *The Astrophysical Journal*, 876(1):85, may 2019. doi: 10.3847/1538-4357/ab1422. URL https://doi.org/10.3847%2F1538-4357%2Fab1422.

K. Schwarzschild. Biography of karl schwarzschild (1873-1916). *Gesammelte Werke Collected Works*, page 1–28, 1992. doi: 10.1007/978-3-642-58086-4_1.

K. S. Stowe. *An introduction to thermodynamics and Statistical Mechanics*. Cambridge Univ. Press, 2013.

L. Susskind. The world as a hologram. *Journal of Mathematical Physics*, 36(11): 6377–6396, 11 1995. ISSN 0022-2488. doi: 10.1063/1.531249. URL https://doi.org/10.1063/1.531249.

M. Tanabashi, K. Hagiwara, K. Hikasa, K. Nakamura, Y. Sumino, F. Takahashi, J. Tanaka, K. Agashe, G. Aielli, C. Amsler, M. Antonelli, D. M. Asner, H. Baer, S. Banerjee, R. M. Barnett, T. Basaglia, C. W. Bauer, J. J. Beatty, V. I. Belousov, J. Beringer, S. Bethke, A. Bettini, H. Bichsel, O. Biebel, K. M. Black, E. Blucher, O. Buchmuller, V. Burkert, M. A. Bychkov, R. N. Cahn, M. Carena, A. Ceccucci, A. Cerri, D. Chakraborty, M.-C. Chen, R. S. Chivukula, G. Cowan, O. Dahl, G. D'Ambrosio, T. Damour, D. de Florian, A. de Gouvêa, T. DeGrand, P. de Jong, G. Dissertori, B. A. Dobrescu, M. D'Onofrio, M. Doser, M. Drees, H. K. Dreiner, D. A. Dwyer, P. Eerola, S. Eidelman, J. Ellis, J. Erler, V. V. Ezhela, W. Fetscher, B. D. Fields, R. Firestone, B. Foster, A. Freitas, H. Gallagher, L. Garren, H.-J. Gerber, G. Gerbier, T. Gershon, Y. Gershtein, T. Gherghetta, A. A. Godizov, M. Goodman, C. Grab, A. V. Gritsan, C. Grojean, D. E. Groom, M. Grünewald, A. Gurtu, T. Gutsche, H. E. Haber, C. Hanhart, S. Hashimoto, Y. Hayato, K. G. Hayes, A. Hebecker, S. Heinemeyer, B. Heltsley, J. J. Hernández-Rey, J. Hisano, A. Höcker, J. Holder, A. Holtkamp, T. Hyodo, K. D. Irwin, K. F. Johnson, M. Kado, M. Karliner, U. F. Katz, S. R. Klein, E. Klempt, R. V. Kowalewski, F. Krauss, M. Kreps, B. Krusche, Y. V. Kuyanov, Y. Kwon, O. Lahav, J. Laiho, J. Lesgourgues, A. Liddle, Z. Ligeti, C.-J. Lin, C. Lippmann, T. M. Liss, L. Littenberg, K. S. Lugovsky, S. B. Lugovsky, A. Lusiani, Y. Makida, F. Maltoni, T. Mannel, A. V. Manohar, W. J. Marciano, A. D. Martin, A. Masoni, J. Matthews, U.-G. Meißner, D. Milstead, R. E. Mitchell, K. Mönig, P. Molaro, F. Moortgat, M. Moskovic, H. Murayama, M. Narain, P. Nason, S. Navas, M. Neubert, P. Nevski, Y. Nir, K. A. Olive, S. Pagan Griso, J. Parsons, C. Patrignani, J. A. Peacock, M. Pennington, S. T. Petcov, V. A. Petrov, E. Pianori, A. Piepke, A. Pomarol, A. Quadt, J. Rademacker, G. Raffelt, B. N. Ratcliff, P. Richardson, A. Ringwald, S. Roesler, S. Rolli, A. Romaniouk, L. J. Rosenberg, J. L. Rosner, G. Rybka, R. A. Ryutin, C. T. Sachrajda, Y. Sakai, G. P. Salam, S. Sarkar, F. Sauli, O. Schneider, K. Scholberg, A. J. Schwartz, D. Scott, V. Sharma, S. R. Sharpe, T. Shutt, M. Silari, T. Sjöstrand, P. Skands, T. Skwarnicki, J. G. Smith, G. F. Smoot, S. Spanier, H. Spieler, C. Spiering, A. Stahl, S. L. Stone, T. Sumiyoshi, M. J. Syphers, K. Terashi, J. Terning, U. Thoma, R. S. Thorne, L. Tiator, M. Titov,

N. P. Tkachenko, N. A. Törnqvist, D. R. Tovey, G. Valencia, R. Van de Water, N. Varelas, G. Venanzoni, L. Verde, M. G. Vincter, P. Vogel, A. Vogt, S. P. Wakely, W. Walkowiak, C. W. Walter, D. Wands, D. R. Ward, M. O. Wascko, G. Weiglein, D. H. Weinberg, E. J. Weinberg, M. White, L. R. Wiencke, S. Willocq, C. G. Wohl, J. Womersley, C. L. Woody, R. L. Workman, W.-M. Yao, G. P. Zeller, O. V. Zenin, R.-Y. Zhu, S.-L. Zhu, F. Zimmermann, P. A. Zyla, J. Anderson, L. Fuller, V. S. Lugovsky, and P. Schaffner. Review of particle physics. *Phys. Rev. D*, 98:030001, Aug 2018. doi: 10.1103/PhysRevD.98.030001. URL https://link.aps.org/doi/10.1103/PhysRevD.98.030001.

E. A. Tropp, V. Y. Frenkel, A. D. Chernin, A. Dron, and M. Burov. *Alexander a friedmann: The man who made the Universe expand*. Cambridge Univ Pr, 2006.

B. Zwiebach. *A first course in string theory*. Cambridge University Press, 2017.